

The Integrals of Gila: What They Are, Which Are Analytic, and How to Finish the Rest

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Abstract

GilaElectromagnetics discretizes the electromagnetic volume integral equation on a lattice of cuboid cells. Every entry of the discretized vacuum Green operator is a sum of four-dimensional integrals of the scalar Helmholtz kernel over pairs of rectangular cell faces. This document explains where those integrals come from, lists exactly which of them Gila now evaluates in closed form and which it still evaluates numerically, proves that a much larger class of them is elementary than the code currently exploits, shows that the remaining ones are not elementary, works out exactly which special functions they reduce to (incomplete Hankel functions for coplanar pairs, apparently a genuinely two-variable function for perpendicular ones) and why the power series in k is nevertheless the right representation in Gila's regime, and collects the techniques, numerical measurements, and pitfalls learned while deriving and hardening the closed forms during the September 2026 work on the `paul-fp32` and `paul-metal` branches. It is written to be self-contained so that the derivation of the remaining closed forms can be taken up in a fresh session.

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1 How to read this document, and context for a fresh session

The repository. The relevant code lives in `src/vacuum/`:

- `glaVacOprMemGen.jl` assembles the Green operator: the face-pair bookkeeping (`facPar`, `srfSum!`, `srfScl`), the fixed Gauss–Legendre rules for separated cells (`egoSrfFxd!`, `quadOrd`, `cntOrd`), the singular-cell corrections (`egoFunSng!`), and the memo of weak integrals (`wekTrp`, `wekMem`).
- `glaVacOprMemInt.jl` holds the scalar kernels (`sclEgo`, `sclEgoN`), the closed-form $1/r$ panel integrals (`rSrfSlf`, `rSrfEdgFlt`, `rSrfEdgCrn`), and the heads of the weakly singular integrals (`wekS`, `wekE`, `wekV` and their `Dir` variants).
- `glaVacOprMemIntSup.jl` is a Julia translation of DIRECTFN [1], the fully numerical scheme for the singular remainders.

Tests for the closed forms are in `test/quadTest.jl` (testset “Panel Integrals”), and the convergence check of the weak integrals is `test/vacuum/intConTest.jl`. The validation scripts of the studies summarized here are kept in `notes/gen/ref/`: `mom.jl` is the high-precision evaluator of every panel-pair moment (Section 6.6), `mom7.jl` the general self-face moment formula with its recurrence and cancellation-free form, `ref.jl` and `formsG.jl` the $1/r$ closed forms, `slf.jl` and `dec.jl` the special-function reduction of the self face, and `moments_report.md` the analysis behind Section 5.2. The generator scripts for the figures, tables and the appendix are in `notes/gen/`.

Conventions. Lengths are measured in wavelengths. The frequency enters through a single complex scalar f , the field `frqPhz` of the kernel options, so that the wavenumber is $k = 2\pi f$. Gila’s scalar kernel is

$$g(r) = \frac{e^{2\pi i f r}}{4\pi f^2 r} = \frac{e^{ikr}}{4\pi f^2 r}, \quad r = |\mathbf{x} - \mathbf{y}|, \quad (1)$$

and its regularized companion, used inside the singular integrals, is $g_N(r) = g(r) - 1/(4\pi f^2 r) = (e^{ikr} - 1)/(4\pi f^2 r)$. A cell has edge lengths $\mathbf{s} = (s_1, s_2, s_3)$, stored as exact `Rationals` (the field `sc1`). When a formula involves a single face I write its edges as l_a, l_b and a third length l_c for the perpendicular direction, matching the argument names of the `rSrf*` functions.

Frequency dependence. Every integral of $1/r$ is proportional to $1/f^2$ and nothing else. This trivial fact turned out to matter (Section 4.5).

2 Where the integrals come from

2.1 From Maxwell to the volume integral equation

In the frequency domain, a scatterer described by a susceptibility χ inside a region Ω produces a total field E satisfying

$$E = E^{\text{inc}} + k^2 \int_{\Omega} \mathbf{G}(\mathbf{x} - \mathbf{y}) \chi(\mathbf{y}) E(\mathbf{y}) d\mathbf{y}, \quad (2)$$

where $\mathbf{G}$ is the dyadic Green function of free space. Gila solves this equation for the polarization current $P = \chi E$, in the form $(\mathbf{I} - \chi \mathbf{G}_0) \cdot = \cdot$ with $\mathbf{G}_0$ the discretized vacuum Green operator, which is why the memory notes refer to the “ $(I - XG_0)$ solve”. The dyadic Green function is

$$\mathbf{G}(\mathbf{r}) = \left(\mathbf{I} + \frac{\nabla \nabla}{k^2} \right) G_s(r), \quad G_s(r) = \frac{e^{ikr}}{4\pi r}, \quad (\nabla^2 + k^2) G_s = -\delta(\mathbf{r}). \quad (3)$$

Using the Helmholtz equation to replace G_s in the identity term, $G_s \mathbf{I} = -(\mathbf{I} \nabla^2 G_s + \mathbf{I} \delta)/k^2$, one gets the form Gila actually uses,

$$\mathbf{G}(\mathbf{r}) = \frac{1}{k^2} (\nabla \nabla - \mathbf{I} \nabla^2) G_s(r) - \frac{\mathbf{I}}{k^2} \delta(\mathbf{r}). \quad (4)$$

Component-wise, with $\partial_a = \partial/\partial x_a$,

$$G_{ab} = \frac{1}{k^2} \left(\partial_a \partial_b - \delta_{ab} \sum_c \partial_c \partial_c \right) G_s - \frac{\delta_{ab}}{k^2} \delta, \quad \text{e.g.} \quad G_{xx} = -\frac{\partial_y \partial_y + \partial_z \partial_z}{k^2} G_s - \frac{\delta}{k^2}, \quad G_{xy} = \frac{\partial_x \partial_y}{k^2} G_s. \quad (5)$$

2.2 Cell averaging and the divergence theorem, twice

Gila uses piecewise-constant basis functions on cuboid cells with Galerkin testing, so the entry of $\mathbf{G}_0$ coupling a source cell V_s to a target cell V_t is the double volume average

$$(\mathbf{G}_0)_{ab}(t, s) = \frac{1}{V_t} \int_{V_t} \int_{V_s} G_{ab}(\mathbf{x} - \mathbf{y}) d\mathbf{y} d\mathbf{x}. \quad (6)$$

The derivatives in (4) are what make this tractable. Because G_s depends on $\mathbf{x} - \mathbf{y}$, a derivative with respect to y_b is minus the derivative with respect to x_b , so $\partial_a \partial_b G_s(\mathbf{x} - \mathbf{y}) = -\partial_a^x \partial_b^y G_s$. Each

derivative can now be moved onto its own integration domain by the divergence theorem, turning a volume integral into a surface integral over the cell boundary:

$$\int_{V_t} \int_{V_s} \partial_a \partial_b G_s \, d\mathbf{y} \, d\mathbf{x} = - \oint_{\partial V_t} \oint_{\partial V_s} n_a(\mathbf{x}) n'_b(\mathbf{y}) G_s(|\mathbf{x} - \mathbf{y}|) \, dS' \, dS, \quad (7)$$

where $\mathbf{n}$ and $\mathbf{n}'$ are the outward normals of the target and source cells. Every second derivative of the singular kernel has become a *surface–surface integral of the scalar kernel itself*, which is only weakly singular ($1/r$ on a four-dimensional domain, integrable). This is the whole reason Gila computes surface integrals: the volume–volume integral of the strongly singular dyadic has been traded for surface–surface integrals of G_s .

For a cuboid cell the boundary consists of six rectangles, each with a constant normal $\pm \mathbf{e}_c$. The double surface integral therefore splits into $6 \times 6 = 36$ *face-pair integrals*

$$I_{FF'} = \int_F \int_{F'} G_s(|\mathbf{x} - \mathbf{y}|) \, dS' \, dS, \quad F \subset \partial V_t, \, F' \subset \partial V_s, \quad (8)$$

and the tensor components are signed sums of them. Writing $\sigma_F = \pm 1$ for the sign of the normal of face F and $c(F)$ for its axis,

$$(\mathbf{G}_0)_{aa} = \frac{1}{k^2 V_t} \sum_{\substack{F, F' \\ c(F)=c(F') \neq a}} \sigma_F \sigma_{F'} I_{FF'} - \frac{\delta_{ts}}{k^2}, \quad (9)$$

$$(\mathbf{G}_0)_{ab} = -\frac{1}{k^2 V_t} \sum_{\substack{F, F' \\ c(F)=a, c(F')=b}} \sigma_F \sigma_{F'} I_{FF'}, \quad a \neq b. \quad (10)$$

Appendix A writes all 36 integrals out with explicit limits, and the nine signed sums. This is exactly what `srfsSum!` does. Its face ordering is `yzL yzU xzL xzU xyL xyU` (faces 1–6, L and U for the lower and upper face on each axis) and the face-pair index is 36-long with `fp = 6(F – 1) + F'`. For example the `xx` entry is `srfsMat[15]–[16]–[21]+[22]+[29]–[30]–[35]+[36]`: pairs 15, 16, 21, 22 are the four combinations of the two y -faces of the target with the two y -faces of the source, with signs $\sigma_F \sigma_{F'}$, and 29–36 likewise for the z -faces. The `xy` entry `–[3]+[4]+[9]–[10]` pairs the x -faces of the target with the y -faces of the source, with the overall minus sign of (7).

The constant prefactors are absorbed into Gila’s kernel (1) and its normalizations. The face-pair integrand is evaluated over the unit hypercube $[0, 1]^4$ and rescaled by `srfsScl`, which is (source face area)/(target edge normal to the target face), i.e. the areas of both faces divided by the target volume. The weak integrals returned by `wekS`, `wekE`, `wekV` are raw integrals $\int_F \int_{F'} g$ over the full rectangles and are divided by the cell volume in `wekTrp`. The identity term $-\delta_{ab} \delta_{ts}/k^2$ appears as the three lines `egoToe[i,i,1,1,1] -= 1/frqPhz^2` in `genEgoCrcS1f!`, which fixes the relation between Gila’s overall constant and the physical one: Gila’s operator equals $4\pi^2$ times (6) in units where the wavelength is one.

2.3 Why so few distinct integrals are needed

On a regular grid the interaction depends only on the relative offset of the two cells, so $\mathbf{G}_0$ is block Toeplitz and is applied with FFTs. Generation therefore needs one 3×3 tensor per relative offset, and each tensor needs 36 face-pair integrals. Moreover the face pairs of offset cells are all axis-aligned rectangle pairs of a small number of *types*, determined by whether the rectangles are coplanar or perpendicular and by how they touch. This is what makes closed forms worth having: a handful of formulas, evaluated once per cell shape, cover every contact interaction in the volume.

2.4 The anti-Hermitian part as a physical check

For real f the panel integrals of $1/r$ are real, and the anti-Hermitian part $\text{asym } \mathbf{G}_0 = (\mathbf{G}_0 - \mathbf{G}_0^\dagger)/2i$ of a Toeplitz operator with an even kernel is the Toeplitz operator of the imaginary part of the kernel. Passivity requires $\text{asym } \mathbf{G}_0 \succeq 0$. This gives a useful end-to-end test of the whole assembly, because it is sensitive only to the imaginary parts and to the frequency prefactors, and it was how the severity of the bug of Section 4.5 became visible: on a 6^3 cell block at $f = 1 + 0.1i$ the buggy operator had $\lambda_{\min}(\text{asym } \mathbf{G}_0) = -0.485$ against $\lambda_{\max} = 0.28$; after the fix all eigenvalues are positive. At real frequency, both before and after, $\lambda_{\min}$ sits at the rounding floor (-5×10^{-15} against $\lambda_{\max} = 0.077$), because the $1/r$ terms do not enter it at all.

3 A taxonomy of the integrals Gila computes

All integrals are of the form (8) with the kernel g of (1). They differ only in the relative position of the two rectangles, and Gila treats them by three methods.

3.1 Contact integrals: rectangles that touch

When the two cells coincide or share a face, an edge, or a vertex, some of their 36 face pairs touch, and the kernel's $1/r$ is on the integration domain. For a regular grid the touching pairs fall into five geometries, illustrated with the source cell $[0, l_a] \times [0, l_b] \times [0, l_c]$:

Self (coincident faces). The same face with itself (self cell), or the upper x -face of the source with the lower x -face of a target displaced by one cell in x : these are the same rectangle. Called **wS** in the code; closed form **rSrfSlf**(l_a, l_b) for the $1/r$ part.

Flat edge (coplanar, sharing an edge). The lower y -faces of two cells adjacent in x : coplanar rectangles $[0, l_a] \times [0, l_c]$ and $[l_a, 2l_a] \times [0, l_c]$ sharing the edge of length l_c . Called **wE** entries 1–6 (**xxY**, **xxZ**, **yyX**, **yyZ**, **zzX**, **zzY**); closed form **rSrfEdgFlt**($l_{\text{shared}}, l_{\text{width}}$), whose second argument is “doubled”: the two panels together span twice its length.

Cornered edge (perpendicular, sharing an edge). Two faces of the same cell that meet at an edge, or the upper y -face of the source with the lower x -face of a target displaced by $(1, 1, 0)$ cells. Called **wE** entries 7–9 (**xy**, **xz**, **yz**); closed form **rSrfEdgCrn**($l_{\text{shared}}, l_b, l_c$). **wekE** computes each of these twice in the two orientations (**xyA**, **xyB**) and averages, which is why the numerical symmetry of the closed form under $l_b \leftrightarrow l_c$ mattered.

Coplanar vertex (coplanar, sharing a corner). The lower z -faces of two cells adjacent along the $(1, 1, 0)$ diagonal. Called **wV** entries 1–3. No closed form; DIRECTFN on the full kernel.

Perpendicular vertex (perpendicular planes, sharing a corner). The upper z -face of the source with the lower x -face of a target displaced by $(1, 1, 1)$ cells. Called **wV** entries 4–6, also averaged over two orientations. No closed form.

Figure 1 draws each geometry with both cells, and Table 1 classifies all 36 face pairs for the four distinct kinds of touching cell pairs (face neighbour, edge neighbour, vertex neighbour, and the cell itself; the other 22 neighbours are these up to reflection). The letter codes are: S coincident, Ef flat edge, Ec cornered edge, Vc coplanar vertex, Vp perpendicular vertex, and for the non-touching pairs of Section 3.2 P1/P2 parallel one or two cell lengths apart, a suffix s for a lateral shift, and Xg perpendicular planes that do not meet. The tables were generated mechanically from the face definitions and agree entry by entry with the correction masks hard-coded in **egoFunSng!** (mask 1 = one of S, Ef, Ec, Vc, Vp; mask 0 = one of P1, P2, P1s, P2s, Xg).

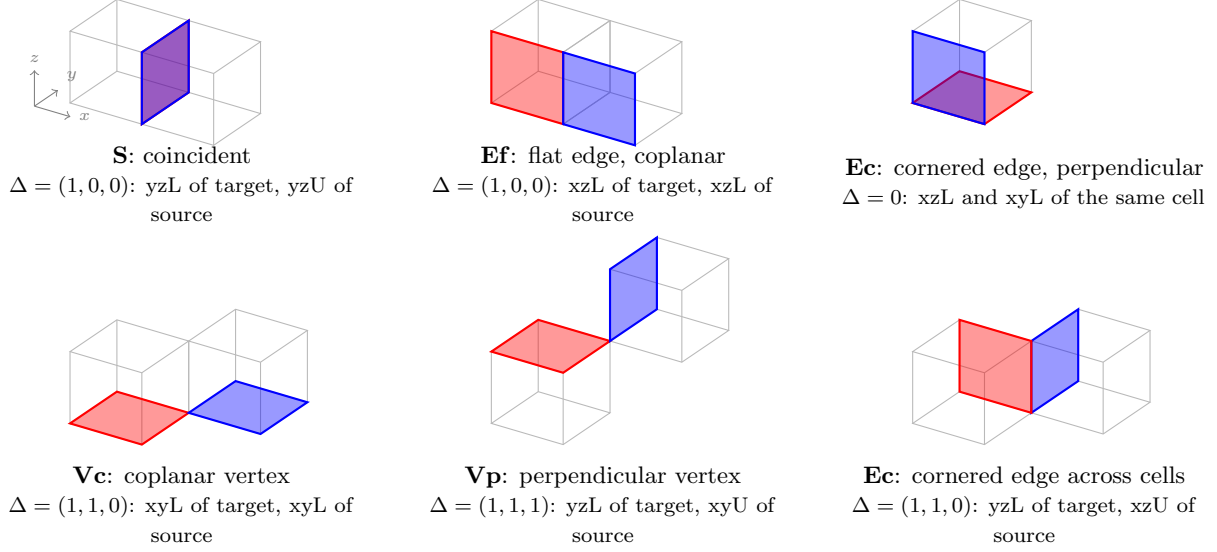


Figure 1: The five contact geometries. Red: source face F' (source cell at the origin); blue: target face F (target cell displaced by Δ cells). Where the two faces coincide (S) the colours overlap. The cornered edge occurs both within one cell and across diagonal neighbours.

$\Delta = (0, 0, 0)$							$\Delta = (1, 0, 0)$						
$F \setminus F'$	yzL	yzU	xzL	xzU	xyL	xyU	$F \setminus F'$	yzL	yzU	xzL	xzU	xyL	xyU
yzL	S	P1	Ec	Ec	Ec	Ec	yzL	P1	S	Ec	Ec	Ec	Ec
yzU	P1	S	Ec	Ec	Ec	Ec	yzU	P2	P1	Xg	Xg	Xg	Xg
xzL	Ec	Ec	S	P1	Ec	Ec	xzL	Xg	Ec	Ec	P1s	Vp	Vp
xzU	Ec	Ec	P1	S	Ec	Ec	xzU	Xg	Ec	P1s	Ec	Vp	Vp
xyL	Ec	Ec	Ec	Ec	S	P1	xyL	Xg	Ec	Vp	Vp	Ec	P1s
xyU	Ec	Ec	Ec	Ec	P1	S	xyU	Xg	Ec	Vp	Vp	P1s	Ec

$\Delta = (1, 1, 0)$							$\Delta = (1, 1, 1)$						
$F \setminus F'$	yzL	yzU	xzL	xzU	xyL	xyU	$F \setminus F'$	yzL	yzU	xzL	xzU	xyL	xyU
yzL	P1s	Ec	Xg	Ec	Vp	Vp	yzL	P1s	Vc	Xg	Vp	Xg	Vp
yzU	P2s	P1s	Xg	Xg	Xg	Xg	yzU	P2s	P1s	Xg	Xg	Xg	Xg
xzL	Xg	Ec	P1s	Ec	Vp	Vp	xzL	Xg	Vp	P1s	Vc	Xg	Vp
xzU	Xg	Xg	P2s	P1s	Xg	Xg	xzU	Xg	Xg	P2s	P1s	Xg	Xg
xyL	Xg	Vp	Xg	Vp	Vc	P1s	xyL	Xg	Vp	Xg	Vp	P1s	Vc
xyU	Xg	Vp	Xg	Vp	P1s	Vc	xyU	Xg	Xg	Xg	Xg	P2s	P1s

Table 1: Geometry of every face pair (F, F') , target face F in rows and source face F' in columns, for the source cell itself and its three kinds of touching neighbour. Codes as in the text.

The numerical method for the first three is a *singularity subtraction*:

$$\iint g = \underbrace{\frac{1}{4\pi f^2} \iint \frac{1}{r}}_{\text{closed form}} + \underbrace{\iint \frac{e^{ikr} - 1}{4\pi f^2 r}}_{\text{DIRECTFN, order intOrd}}. \quad (11)$$

The regularized kernel g_N is bounded (it tends to $ik/(4\pi f^2) = i/(2f)$ as $r \rightarrow 0$; the code evaluates it as $i \text{cispi}(fr) \text{sinc}(fr)/(2f)$ to avoid cancellation, see Section 4.6). DIRECTFN then integrates it by splitting each rectangle into two triangles and applying a chain of variable transformations that map the four-dimensional triangle-pair integral onto a smooth integrand over a box, which is integrated by tensor Gauss–Legendre rules with `intOrd` points per dimension. The default is now `intOrd = 48`, so a self-face costs of order $48^4 \approx 5 \times 10^6$ kernel evaluations. The vertex integrals use DIRECTFN on the *unregularized* kernel (the flag `sngMod = false` in `wekVint` selects `kerEV`, the full g), because the vertex singularity is mild enough for the transformations alone.

3.2 Touching-shell integrals: rectangles of touching cells that do not touch

For a pair of touching cells most of the 36 face pairs do *not* touch: they are at least one cell length apart. Two faces of touching cells with the same normal axis either lie in the same plane, in which case they are one of the contact geometries above (coplanar faces of touching cells always meet, so a coplanar pair with a gap never occurs on the lattice), or lie in parallel planes one or two cell lengths apart, possibly shifted laterally by a cell length when the cells are diagonal neighbours. Faces with different normal axes either share an edge or a vertex (contact) or sit in perpendicular planes a cell length apart without meeting. These integrals are regular but nearly singular, and Gila integrates them with a fixed Gauss–Legendre tensor rule of order `cntOrd = 9` (`pairListUn` in `egoFunSng!`; the mask tables there mark them with a 0). Figure 2 draws the five cases; their counts per neighbour type can be read off Table 1: 6 of 36 for the cell itself (all P1), 15 for a face neighbour, 22 for an edge neighbour, 27 for a vertex neighbour.

3.3 Separated integrals: cells two or more apart

For cells whose max-norm separation is at least two cells, every face pair is separated by at least one cell length, and the integrand is smooth. Gila uses a fixed tensor Gauss–Legendre rule whose order comes from the measured schedule in `quadOrd`: 9 points per dimension at 2 cells, 7 at 3–4, 6 at 5–6, 5 at 7–16, 4 beyond, with a floor of 5 for cells above $\lambda/8$ and 6 above $\lambda/4$. The schedule targets 10^{-10} relative error on the assembled tensor with one order of margin. It was measured, not derived, and is described in the comment above `quadOrd`. The measurement also showed why assembled accuracy is the right target: `srfSum!` subtracts face pairs of nearly equal size, amplifying face-pair error by a factor between 10 and 1000, so a face pair must be good to 10^{-12} – 10^{-13} for the tensor to be good to 10^{-10} .

Cross-scale volume pairs whose cells are not aligned to a common lattice fall through to adaptive cubature (`egoSrfAdp!`, `HCubature`); these are rare and not discussed further.

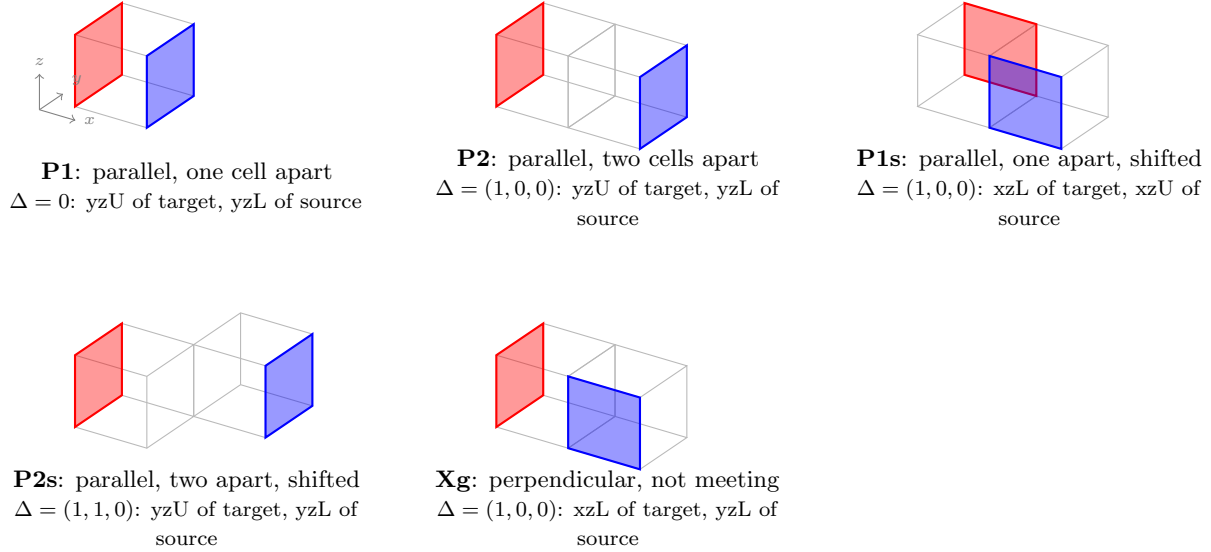


Figure 2: The five non-touching face-pair geometries that occur between touching cells. Same colour convention as Figure 1.

3.4 Summary table

Geometry	Kernel part	Method today	Analytic?	Analytic possible?
Self face	$1/r$	closed form rSrfS1f	yes	—
Self face	$(e^{ikr} - 1)/r$	DIRECTFN, order 48	no	yes, series (Sec. 5.1, 5.2)
Flat edge	$1/r$	closed form rSrfEdgFlt	yes	—
Flat edge	$(e^{ikr} - 1)/r$	DIRECTFN	no	yes, series
Cornered edge	$1/r$	closed form rSrfEdgCrn	yes	—
Cornered edge	$(e^{ikr} - 1)/r$	DIRECTFN	no	yes, series
Coplanar vertex	e^{ikr}/r	DIRECTFN, full kernel	no	yes: $1/r$ closed form + series
Perpendicular vertex	e^{ikr}/r	DIRECTFN, full kernel	no	yes: $1/r$ closed form + series
Parallel offset $1\times, 2\times$	e^{ikr}/r	GL order 9	no	yes, series (measured)
Parallel, laterally shifted	e^{ikr}/r	GL order 9	no	yes, series (expected)
Perpendicular, not meeting	e^{ikr}/r	GL order 9	no	yes, series (expected)
Separated ≥ 2 cells	e^{ikr}/r	GL order 4–9	no	not usefully (Sec. 5.5)

4 What has been done analytically

Only the $1/r$ parts of the self, flat-edge and cornered-edge integrals are closed form. These three formulas existed in Gila from its first commit as Mathematica-generated expressions of 20–30 terms in logarithms, square roots, and inverse trigonometric functions. In September 2026 they were replaced by short equivalent forms derived by hand, for two reasons: one of them was wrong for complex frequency, and all three lost most of their digits for slender panels. This section records the derivations, since they are the templates for what remains.

4.1 The difference-variable reduction

The central trick for coplanar rectangles is to integrate over the *difference* of the two points rather than the points themselves. Let the target panel be $A = [0, p]$ in one coordinate and the source panel $B = [c, c + p']$. The kernel depends only on $u = x' - x$. For fixed u , the measure of pairs $(x, x') \in A \times B$ with $x' - x = u$ is the convolution of the two indicator functions, a trapezoid in u : for two equal intervals with the same origin it is the triangle $p - |u|$ on $[-p, p]$; for adjacent intervals $[0, p]$ and $[p, 2p]$ it is the triangle $p - |u - p|$ centred at $u = p$; for an interval against a degenerate one (a point, when the other rectangle lies in a perpendicular plane) it is an indicator. Doing this in both in-plane coordinates reduces the four-dimensional integral to a two-dimensional one with a piecewise-bilinear weight, and by symmetry in the signs of u and v ,

$$\int_{A \times A} \int_{B \times B} \frac{dS dS'}{|\mathbf{x} - \mathbf{y}|} = 4 \int_0^p \int_0^q \frac{(p-u)(q-v)}{\sqrt{u^2 + v^2}} dv du \quad (\text{self panel } p \times q). \quad (12)$$

Expanding the weight gives four *moments* of $1/\sqrt{u^2 + v^2}$ over the rectangle,

$$M_{ij}(p, q) = \int_0^p \int_0^q \frac{u^i v^j}{\sqrt{u^2 + v^2}} dv du, \quad \iint \frac{1}{r} = 4[pq M_{00} - q M_{10} - p M_{01} + M_{11}]. \quad (13)$$

Each moment is elementary. In polar coordinates $u = \rho \cos \theta$, $v = \rho \sin \theta$ the $1/r = 1/\rho$ cancels the Jacobian ρ , the radial integral of a monomial in ρ is a power of the boundary radius $p/\cos \theta$ or $q/\sin \theta$, and the angular integrals of powers of $\sec \theta$ and $\csc \theta$ are elementary. With $h = \sqrt{p^2 + q^2}$:

$$\begin{aligned} M_{00} &= p \operatorname{asinh}(q/p) + q \operatorname{asinh}(p/q), & M_{10} &= \frac{p^2}{2} \operatorname{asinh}(q/p) + \frac{q}{2}(h - q), \\ M_{01} &= \frac{q^2}{2} \operatorname{asinh}(p/q) + \frac{p}{2}(h - p), & M_{11} &= \frac{1}{3}(h^3 - p^3 - q^3). \end{aligned} \quad (14)$$

4.2 The self panel: thirty terms become three

Substituting (14) into the bracket, the coefficients of the two asinh terms combine to $\frac{1}{2}p^2q$ and $\frac{1}{2}pq^2$, and the algebraic terms combine, using $h(p^2 + q^2) = h^3$, to $(p^3 + q^3 - h^3)/6$. Including the $1/(4\pi f^2)$ of the kernel,

$$\boxed{\mathbf{rSrfsIf}(l_a, l_b) = \frac{l_a^3 + l_b^3 - h^3 + 3l_a^2 l_b \operatorname{asinh}(l_b/l_a) + 3l_a l_b^2 \operatorname{asinh}(l_a/l_b)}{6\pi f^2}, \quad h = \sqrt{l_a^2 + l_b^2}. \quad (15)}$$

This is identical to the original thirty-term expression: checked in 512-bit arithmetic to 10^{-148} relative at aspect ratios from 10^{-4} to 10^4 , and Mathematica's `FullSimplify` of the difference leaves a residue of logarithms that vanishes by $(h - l_a)(h + l_a) = l_b^2$. The original contained $\sqrt{l_a^2 + 4l_b^2}$ throughout; a self integral cannot depend on a doubled edge, and indeed every such term cancels. The cubic difference is evaluated cancellation-free as

$$l_a^3 + l_b^3 - h^3 = p^3 - p^2 \frac{h^2 + hq + q^2}{h + q}, \quad p = \min(l_a, l_b), \quad q = \max(l_a, l_b), \quad (16)$$

which follows from $h^3 - q^3 = (h - q)(h^2 + hq + q^2)$ and $h - q = p^2/(h + q)$.

4.3 The flat edge pair

The target is $[0, l_a] \times [0, l_b]$ and the source $[0, l_a] \times [l_b, 2l_b]$. In the v direction the convolution weight is the triangle $l_b - |v - l_b|$ on $(0, 2l_b)$. Splitting at $v = l_b$ gives an integral with weight $(l_a - u)v$ over $[0, l_a] \times [0, l_b]$ and one with weight $(l_a - u)(2l_b - v)$ over $[0, l_a] \times [l_b, 2l_b]$; the second is a difference of rectangles of heights $2l_b$ and l_b . Everything is again the moments (14), and collecting terms with $h_1 = \sqrt{l_a^2 + l_b^2}$, $h_2 = \sqrt{l_a^2 + 4l_b^2}$,

$$\boxed{\text{rSrfEdgFlt}(l_a, l_b) = \frac{Q + 6l_a^2 l_b D_1 + 6l_a l_b^2 D_2}{12\pi f^2}} \quad (17)$$

with

$$Q = 2h_1^3 - h_2^3 - l_a^3 + 6l_b^3, \quad D_1 = \text{asinh} \frac{2l_b}{l_a} - \text{asinh} \frac{l_b}{l_a}, \quad D_2 = 2 \text{asinh} \frac{l_a}{2l_b} - \text{asinh} \frac{l_a}{l_b}. \quad (18)$$

All three pieces cancel catastrophically as written when $l_a \ll l_b$ (the shared edge much shorter than the panel width): Q is $O(l_a^3 l_b^2 / l_b^3)$ out of terms of size l_b^3 , and the asinh differences are small differences of large logarithms. The implementation writes Q as a manifestly negative product,

$$Q = -2l_a^3 l_b^2 \left[\frac{3l_a}{(h_2 + 2h_1)(h_1 + l_b)(h_2 + 2l_b)} + \frac{1 + 3l_a/(h_2 + 2h_1)}{(h_1 + l_a)(h_2 + l_a)} \right], \quad (19)$$

and the differences as $\log(1 + x)$ of positive ratios, $D_1 = \log(1 + (l_b + 3l_b^2/(h_1 + h_2))/(l_b + h_1))$ and $D_2 = \log(1 + l_a^3 [l_a/(h_1 + l_b)^2 + 1/(h_2 + 2l_b)] / (2l_b(l_a + h_1)))$, using $\text{asinh} x = \log(x + \sqrt{1 + x^2})$ and rationalizing every difference of a square root and a length. The result loses no measurable digits over nine decades of aspect ratio, where the original lost up to fifteen.

4.4 The cornered edge pair and the box potential

Perpendicular rectangles do not reduce to a two-dimensional difference variable, because the two in-plane coordinates of one panel are not the in-plane coordinates of the other. Take the shared edge along $u \in [0, l_a]$, one panel in the plane $z = 0$ extending to $y \in [0, l_b]$, the other in the plane $y = 0$ extending to $z \in [0, l_c]$. Only the shared direction admits a difference variable (weight $l_a - u$), and the other two coordinates stay as they are:

$$\iint \frac{1}{r} = 2 \int_0^{l_a} (l_a - u) \int_0^{l_b} \int_0^{l_c} \frac{dz dy}{\sqrt{u^2 + y^2 + z^2}} du. \quad (20)$$

The inner double integral is the Newtonian potential of a rectangle $[0, l_b] \times [0, l_c]$ at a point at height u above its corner, which is elementary; integrating by parts in u turns the weight $l_a - u$ into an antiderivative and gives

$$\iint \frac{1}{r} = 2 \int_0^{l_a} N_0(u, l_b, l_c) du, \quad (21)$$

where $N_0(a, b, c) = \int_0^a \int_0^b \int_0^c dx dy dz / \sqrt{x^2 + y^2 + z^2}$ is the potential of a box at its own corner, a classical result:

$$\begin{aligned} N_0(a, b, c) = & ab \text{asinh} \frac{c}{\sqrt{a^2 + b^2}} + bc \text{asinh} \frac{a}{\sqrt{b^2 + c^2}} + ca \text{asinh} \frac{b}{\sqrt{c^2 + a^2}} \\ & - \frac{a^2}{2} \text{atan} \frac{bc}{aR} - \frac{b^2}{2} \text{atan} \frac{ca}{bR} - \frac{c^2}{2} \text{atan} \frac{ab}{cR}, \end{aligned} \quad (22)$$

with $R = \sqrt{a^2 + b^2 + c^2}$. Its remaining u -integral is elementary, and after grouping (with $h_{ab} = \sqrt{l_a^2 + l_b^2}$ etc.)

$$\boxed{\text{rSrfEdgCrn} = \frac{J}{2\pi f^2}}, \quad J = \frac{l_b^3}{6}\Delta_b + \frac{l_c^3}{6}\Delta_c + \frac{l_a^2 l_b}{2} \operatorname{asinh} \frac{l_c}{h_{ab}} + \frac{l_a^2 l_c}{2} \operatorname{asinh} \frac{l_b}{h_{ac}} + l_a l_b l_c \operatorname{asinh} \frac{l_a}{h_{bc}} - \frac{l_a^3}{6} \operatorname{atan} \frac{l_b l_c}{l_a R} - \frac{l_a l_b^2}{2} \operatorname{atan} \frac{l_a l_c}{l_b R} - \frac{l_a l_c^2}{2} \operatorname{atan} \frac{l_a l_b}{l_c R} - \frac{l_a^2 l_b l_c}{3(h_{bc} + R)}, \quad (23)$$

where $\Delta_b = \operatorname{asinh}(l_c/l_b) - \operatorname{asinh}(l_c/h_{ab})$ and Δ_c is its mirror. Grouping the l_b^3 and l_c^3 logarithms into these differences was the essential step: without it the form still lost 14.6 digits when $l_a \ll l_b \ll l_c$. Written this way, and with the differences as $\log(1+x)$ of positive ratios, the worst loss over the 9×9 grid of aspect ratios 10^{-4} to 10^4 in both l_b/l_a and l_c/l_a is 0.4 digits, against 18.8 for the original. The form is manifestly symmetric in $l_b \leftrightarrow l_c$, so the two orientations that **wekE** averages now agree to 6×10^{-16} .

4.5 The bug: a missing $1/f^2$

The original **rSrfEdgFlt** consisted of two blocks, one with prefactor $1/(12\pi f^2)$ and one with prefactor $1/(64\pi)$. Since the integrand is $1/(4\pi f^2 r)$, every term must scale as $1/f^2$; the second block did not. At $f = 1$ the function was correct (it matched an independent evaluation to 10^{-16}), so nothing in the real-frequency tests could see the error, but at $f = 1 + 0.1i$ the flat-edge terms were off by 23%, the assembled operator by 20%, and the anti-Hermitian part was indefinite (Section 2.4). The lesson generalizes: *every closed form should be tested for its homogeneity in f* , which is a one-line test, and complex frequency should be part of any test that claims to cover frequency dependence. The new form has a single prefactor.

4.6 The scalar kernels

Two related evaluation changes were made. The regularized kernel is computed as

$$g_N(r) = \frac{e^{ikr} - 1}{4\pi f^2 r} = \frac{i e^{i\pi x} \operatorname{sinc}(x)}{2f}, \quad x = fr, \quad \operatorname{sinc}(x) = \frac{\sin \pi x}{\pi x}, \quad (24)$$

from the half-angle identity $e^{iz} - 1 = 2i \sin(z/2) e^{iz/2}$ with $z = 2\pi x$. There is no subtraction, sinc is entire so $r = 0$ needs no branch, and Julia's **cispi** and **sinc** reduce the argument modulo π exactly. Against a 256-bit reference the form is accurate to about two units in the last place for real and complex f ; the previous $\operatorname{cis}(2\pi x) - 1$ with a Taylor branch below $r = 10^{-7}$ lost up to six digits near the switch. The far kernel uses **cispi**($2fr$) instead of **cis**($2\pi fr$), which avoids rounding the product $2\pi fr$ before argument reduction and is one to two orders of magnitude more accurate at separations of ten wavelengths and beyond. Past that, the far-kernel error is set by the rounding of r itself (a relative ϵ in r is an absolute phase error $2\pi fr \epsilon$), which no evaluation of the exponential can remove.

4.7 How the closed forms were verified

The verification protocol is worth keeping, because it caught real problems:

1. **Equivalence to the original formula in extended precision.** Copy the original into a generic function, and evaluate in **BigFloat** at 256–512 bits. *Pitfall:* in Julia, `48 * pi`, `log(64)`, `log(2)` evaluate to **Float64** before they meet a **BigFloat**; every constant must be typed to the working precision (`oftype(la, pi)`, `log(64 * one(la))`) or the reference is silently a **Float64** one and the comparison proves nothing.

2. **Digits lost.** Report $\log_{10}(|x_{64} - x_{\text{big}}|/(\epsilon|x_{\text{big}}|))$ on a grid of aspect ratios. When the same digit count is lost at 256 bits as at 53, the loss is an intrinsic condition number of the expression as written, not an evaluation-order artefact, and only regrouping helps.
3. **Independent value at $f = 1$.** For the moments this was the exact polar-coordinate closed form; for the cornered pair a one-dimensional adaptive integral of N_0 in `BigFloat`; both were cross-checked against `HCubature` and a hand-evaluated unit-square integral ($\int_{[0,1]^2 \times [0,1]^2} d/r = 2.9732095982473794\dots$).
4. **Homogeneity in f ,** symmetry in $l_b \leftrightarrow l_c$, and the anti-Hermitian positivity check of Section 2.4.
5. **Comparison with Gila's own numerics at two orders.** The `intOrd-48` and `intOrd-64` values of `wek*` converge monotonically toward the closed-form-plus-series value (Section 5.4), which is the strongest confirmation that the geometry bookkeeping of a new formula matches the code.

5 What has not been done, and whether it can be

Not analytic today: the $1/r$ parts of both vertex geometries; the regular remainders $(e^{ikr} - 1)/r$ of all five contact geometries; the full kernel on the three touching-shell geometries; and the full kernel on separated pairs. The rest of this section shows that everything except the last is analytic in a precise sense, and that the last is not.

5.1 Every power of r is elementary

The key observation is that the full kernel is an entire function of kr ,

$$g(r) = \frac{1}{4\pi f^2} \sum_{n=0}^{\infty} \frac{(ik)^n}{n!} r^{n-1}, \quad (25)$$

so a face-pair integral of g is the same series in the *moments*

$$I_m(F, F') = \int_F \int_{F'} |\mathbf{x} - \mathbf{y}|^m dS' dS, \quad m = -1, 0, 1, 2, \dots \quad (26)$$

For even m the integrand is a polynomial in the coordinates and I_m is a trivial exact rational combination of the edge lengths. The content is in odd m .

Theorem 5.1 (Elementarity of the moments). *For axis-aligned rectangles $F, F' \subset \mathbb{R}^3$ and every integer $m \geq -1$, $I_m(F, F')$ is an elementary function of the edge lengths and offsets: a finite combination of polynomials, square roots of sums of squares, asinh (equivalently logarithms), and atan .*

Proof sketch, constructive. Write the pair of points as a single point $X = (\mathbf{x}, \mathbf{y}) \in \mathbb{R}^6$ ranging over the product $P = F \times F'$, which is a four-dimensional axis-aligned box (a rectangle times a rectangle, embedded in $\mathbb{R}^6$ with two coordinates frozen). The distance is $r = |AX|$ with $A = [\mathbf{I} \quad -\mathbf{I}]$ the linear map $(\mathbf{x}, \mathbf{y}) \mapsto \mathbf{x} - \mathbf{y}$. For any linear A and any m , the vector field $X |AX|^m$ has divergence

$$\nabla \cdot (X |AX|^m) = N |AX|^m + m |AX|^{m-2} X \cdot A^\top AX = (N + m) |AX|^m \quad (27)$$

in N dimensions, since $X \cdot A^\top AX = |AX|^2$. Applying the divergence theorem on the box P (here $N = 4$, restricting X to the four free coordinates),

$$\int_P |AX|^m dX = \frac{1}{N + m} \sum_{\text{facets } Q \text{ of } P} (X_Q \cdot \nu_Q) \int_Q |AX|^m dS, \quad (28)$$

where ν_Q is the outward normal of the facet and $X_Q \cdot \nu_Q$ is constant on the facet (the coordinate of the facet's plane). Each facet is a three-dimensional box, on which $|AX|^2$ restricted is a quadratic form in the facet coordinates plus a constant: completing the square, $|AX|^2 = |BY|^2 + c^2$ with Y the facet coordinates measured from the foot of the perpendicular, B the restriction of A , and c the distance from the origin of A 's range to the facet's affine image. If B has a null direction (both rectangles extend along the same axis, so a facet coordinate does not change $\mathbf{x} - \mathbf{y}$), the integral along that direction is a length factor and the dimension drops for free. Otherwise the same identity with the shifted kernel gives

$$\nabla \cdot (Y (|BY|^2 + c^2)^{m/2}) = (N' + m) (|BY|^2 + c^2)^{m/2} - m c^2 (|BY|^2 + c^2)^{(m-2)/2}, \quad (29)$$

so the facet integral of exponent m is a boundary sum of exponent m over $(N' - 1)$ -dimensional faces plus c^2 times a facet integral of exponent $m - 2$. The recursion lowers the dimension and the exponent together and terminates in one-dimensional integrals $\int (\beta^2 t^2 + c^2)^{m'/2} dt$ over intervals, which for every integer m' are elementary: polynomials for even $m' \geq 0$, asinh plus algebraic terms for odd m' , algebraic for negative odd m' , and atan for $m' = -2$. The exceptional prefactors $1/(N' + m)$ with $N' + m = 0$ occur only when a lower-dimensional face is reached with $m = -N'$, which produces the logarithm and arctangent terms rather than a genuine obstruction; and the singularity of r^{-1} on the set $AX = 0$ (rectangles that touch) is integrable and handled by excising a shrinking neighbourhood, exactly as in the classical proofs for the Newtonian potential of polyhedra [2, 3]. $\square$

The theorem is not new in substance (the Newtonian potential of a homogeneous polyhedron is the case $m = -1$, $N = 3$, and integrals of r^m over polytopes appear in the mass-property and view-factor literature), but its statement for *pairs* of rectangles via the product box seems to be the useful packaging: it handles the coplanar and perpendicular geometries, and the weights $(p-u)(q-v)$ of the difference-variable reduction, in one mechanism, because the weights are nothing but the geometry of the product box's facets.

Corollary 5.2. *The $1/r$ parts of both vertex geometries are elementary, and so are all the odd moments $I_1, I_3, I_5, \dots$ of all eight contact and touching-shell geometries. In particular, the full kernel on every touching or one-cell-apart face pair is an absolutely convergent series of elementary terms.*

5.2 The self-panel moments for every m : a worked example

The theorem is abstract; here is what it produces in the simplest case, worked out from Mathematica's output for $m = 0, \dots, 6$ of

$$\text{mom}(m) = \int_0^p \int_0^q (p-u)(q-v) (u^2 + v^2)^{m+1/2} dv du, \quad I_{2m+1}(\text{self}) = 4 \text{mom}(m), \quad (30)$$

and then derived for general m in polar coordinates. Mathematica's expressions are correct (checked to 10^{-65} at 220 bits) but hide their structure: they present an even polynomial divided by h and scatter the logarithms, and the $m = 0, 1, 2$ outputs contain log combinations that are identically small asinh values (the $m = 1$ combination $-15 \text{asinh } \frac{p}{q} + 5 \log q - 12 \log(h-p) + 7 \log(p+h)$ is exactly $4 \text{asinh}(p/q)$), so as printed they lose *all* significance at aspect ratio 10^{-8} . Done by hand

the pattern is clean. With $h = \sqrt{p^2 + q^2}$, for every integer $m \geq -1$,

$$\text{mom}(m) = B_m \left[p^2 q^2 \sum_{j=1}^{m+1} \frac{(2j-2)!!}{(2j-1)!!} h^{2j-1} (p^{2(m+1-j)} + q^{2(m+1-j)}) + p q^{2m+4} \operatorname{asinh} \frac{p}{q} + q p^{2m+4} \operatorname{asinh} \frac{q}{p} \right] - \frac{h^{2m+5} - p^{2m+5} - q^{2m+5}}{(2m+3)(2m+4)(2m+5)}, \quad B_m = \frac{(2m+1)!!}{(2m+2)!!(2m+3)(2m+4)}, \quad (31)$$

At $m_3 = -1$ the sum is empty and the formula is the $1/r$ self integral (15): the cubic difference $h^3 - p^3 - q^3$ is the first member of the family $h^{2m+5} - p^{2m+5} - q^{2m+5}$, and the two asinh terms carry $p^2 q$ and $p q^2$. So one formula covers the whole series for the self face, and no separate derivation of the $1/r$ case was ever needed.

Derivation. In polar coordinates about the corner the radial integral is exact,

$$\text{mom}(m) = \int_0^{\pi/2} \left[\frac{pq R^N}{N} - \frac{(p \sin \theta + q \cos \theta) R^{N+1}}{N+1} + \frac{\cos \theta \sin \theta R^{N+2}}{N+2} \right] d\theta, \quad N = 2m+3, \quad (32)$$

with $R = p \sec \theta$ on $[0, \theta_0]$ and $R = q \csc \theta$ on $[\theta_0, \pi/2]$, $\theta_0 = \arctan(q/p)$. On the first wedge the $\sin \theta \sec^{N+1} \theta$ pieces integrate directly, since $d(\sec^N \theta)/d\theta = N \sec^N \theta \tan \theta$, and give the $h^{2m+5} - p^{2m+5}$ family; the $\sec^N \theta$ pieces combine to $\frac{qp^{N+1}}{N(N+1)} S_N$ with $S_N = \int_0^{\theta_0} \sec^N \theta d\theta$, and the reduction $\int \sec^N = \frac{\sec^{N-2} \tan}{N-1} + \frac{N-2}{N-1} \int \sec^{N-2}$ with $\sec \theta_0 = h/p$, $\tan \theta_0 = q/p$, $S_1 = \operatorname{asinh}(q/p)$ produces the double-factorial coefficients. The second wedge is the $p \leftrightarrow q$ mirror. The same computation gives a two-term recurrence that generates all m at constant cost: with $X_m(p, q) = q p^{2m+4} S_{2m+3}$,

$$X_{-1} = p^2 q \operatorname{asinh} \frac{q}{p}, \quad X_m = \frac{p^2 q^2 h^{2m+1}}{2m+2} + \frac{2m+1}{2m+2} p^2 X_{m-1}, \quad (33)$$

$$\text{mom}(m) = \frac{X_m(p, q) + X_m(q, p)}{(2m+3)(2m+4)} - \frac{h^{2m+5} - p^{2m+5} - q^{2m+5}}{(2m+3)(2m+4)(2m+5)}. \quad (34)$$

Verification. Against Mathematica for $m = 0..6$: 1.4×10^{-65} . Against the independent 220-bit evaluator of Section 6.6 for $m = 0..10$ at aspect ratios up to 10^3 : 1.3×10^{-40} , which is the evaluator's own refinement residual. Recurrence against closed form: 7×10^{-66} .

Cancellation. As $p \rightarrow 0$, $\text{mom}(m) \rightarrow p^2 q^{2m+3} / (2(2m+2)(2m+3))$. Every group in (31) is already of that size except $h^{2m+5} - q^{2m+5}$, a difference of two $O(q^{2m+5})$ numbers that costs $2 \log_{10}(q/p)$ digits. Writing it as the product

$$h^n - b^n = \frac{a^2}{h+b} \sum_{i=0}^{n-1} h^i b^{n-1-i}, \quad n = 2m+5, \quad (a, b) = \min \max(p, q), \quad (35)$$

and subtracting the negligible a^n leaves an expression in which every remaining term is positive. Digits lost in Float64, worst over $m \leq 10$:

p/q	10^{-8}	10^{-4}	10^{-2}	1	10^2	10^4	10^8
Mathematica as printed	14.7	6.2	2.9	0.1	3.2	7.3	all
(31) as written	15.6	7.3	3.9	0.6	3.1	7.8	15.6
regrouped	0.5	0.2	0.5	0.6	0.4	0.7	0.5

The first fully closed-form Gila contact integral. With I_{-1} and all odd moments from (31), and the even moments as the exact polynomial

$$I_{2j}(\text{self}) = 4 \sum_{k=0}^j \binom{j}{k} \frac{p^{2k+2}}{(2k+1)(2k+2)} \frac{q^{2(j-k)+2}}{(2j-2k+1)(2j-2k+2)}, \quad (36)$$

the series (25) for the self face was summed in `Float64` and compared with `wekS` at `intOrd` 64:

cell	face (p, q)	relative	real part	imaginary part
$(1/32)^3$	$(1/32, 1/32)$	8.1×10^{-13}	8.1×10^{-13}	1.3×10^{-15}
$(1/8)^3$	$(1/8, 1/8)$	1.3×10^{-11}	1.4×10^{-11}	1.1×10^{-15}
$(1/32, 1/32, 1/512)$	$(1/32, 1/512)$	2.6×10^{-12}	2.6×10^{-12}	7.8×10^{-16}

The series is not the limit: its `Float64` evaluation agrees with a 300-bit one to 1.2×10^{-16} , and truncating at 40 or 60 terms gives bitwise identical results. The whole discrepancy is `DIRECTFN`'s, matching the earlier estimate of its order-64 error. The split between real and imaginary parts is diagnostic: the imaginary part of g is $\sin(kr)/r$, an entire function of r^2 , so it is carried entirely by the even moments, which are polynomials that any quadrature integrates well; the real part contains $1/r$ and the odd powers $r, r^3, \dots$, which are the pieces with a kink at $r = 0$, and that is exactly where Gila's remaining contact-integral error lives.

5.3 The full kernel: not elementary, but what special functions?

Could the face-pair integral of e^{ikr}/r itself be written in closed form, sparing us the series? Not in elementary functions, and the interesting question is what *special* functions appear and whether they are ones we can evaluate. This was worked out completely for the self panel and verified in 220–300-bit arithmetic against the four-dimensional evaluator of Section 6.6.

Exact reduction of the self panel. Start from the difference-variable form $I = 4 \int_0^p \int_0^q (p - u)(q - v) e^{ik\rho}/\rho$. In polar coordinates the $1/\rho$ cancels the Jacobian and the radial integrand is a polynomial times $e^{ik\rho}$, so the radial integrals are elementary in $e^{ikR(\theta)}$ with $R = p \sec \theta$ or $q \csc \theta$. A pleasant surprise: every $\sec^j \theta$ with $j \geq 1$ cancels identically, because the bilinear weight vanishes on the far edge of each sector (the convolution weight is zero at the boundary of its support). What remains is

$$I(k) = 4 \left[\frac{i\pi pq}{2k} + \frac{p+q}{k^2} - \frac{i}{k^3} \right] + S(p, q) + S(q, p), \quad (37)$$

$$S(p, q) = 4 \int_0^{\arctan(q/p)} e^{ikp \sec \theta} \left[-\frac{q \cos \theta - p \sin \theta}{k^2} + \frac{2i \sin \theta \cos \theta}{k^3} \right] d\theta. \quad (38)$$

Verified against the four-dimensional value to 6×10^{-60} for cells of $\lambda/32$ and $\lambda/4$, real and complex k , and to 4×10^{-44} for the 16:1 slender panel, where the residual is the four-dimensional evaluator's own error.

Naming the pieces. Substitute $y = \sec \theta = \cosh t$, $z = kp$, $Y = D/p$ with $D = \sqrt{p^2 + q^2}$, $T = \operatorname{arccosh} Y$, and define

$$F_m(z, Y) = \int_1^Y \frac{e^{izy}}{y^m \sqrt{y^2 - 1}} dy = \int_0^T e^{iz \cosh t} \operatorname{sech}^m t dt, \quad P_m(z, Y) = \int_1^Y e^{izy} y^{-m} dy. \quad (39)$$

The $\sin \theta$ -weighted pieces are the elementary-in-structure P_m , and the $\cos \theta$ piece is F_2 :

$$S(p, q) = 4 \left[-\frac{q}{k^2} F_2(kp, \frac{D}{p}) + \frac{p}{k^2} P_2(kp, \frac{D}{p}) + \frac{2i}{k^3} P_3(kp, \frac{D}{p}) \right], \quad (40)$$

$$P_m(z, Y) = E_m(-iz) - Y^{1-m} E_m(-izY), \quad E_m(x) = x^{m-1} \Gamma(1-m, x), \quad (41)$$

both verified to 10^{-64} . The E_m are generalized exponential integrals, i.e. incomplete gamma functions, and are unproblematic. So the self-panel integral is: elementary terms, three exponential integrals (at $-ikp$, $-ikq$, $-ikD$), and *two evaluations of the single two-argument function F_2* .

What F_2 is. It belongs to the incomplete Hankel family. With the incomplete Hankel function defined by truncating the Sommerfeld integral,

$$H_\nu^{(1)}(z, w) = \frac{2}{i\pi} e^{-i\nu\pi/2} \int_0^w e^{iz \cosh t} \cosh(\nu t) dt \xrightarrow{w \rightarrow \infty} H_\nu^{(1)}(z) \quad (42)$$

(normalization checked numerically against the complete Hankel functions for $\nu = 0, 1, 2$), one has $F_0 = \frac{i\pi}{2} H_0^{(1)}(z, T)$ and $F_{-1} = -\frac{\pi}{2} H_1^{(1)}(z, T)$. Because $\cosh(\nu \operatorname{arccosh} y)$ is a polynomial in y , the $\cosh(\nu t)$ family covers exactly the F_m with $m \leq 0$; our F_2 has a *negative* power of y and lies outside it. It is reached by a verified recursion and by integration in z :

$$F_2(z, Y) = \frac{\sqrt{Y^2 - 1}}{Y} e^{izY} + iz[F_1 - F_{-1}], \quad F_2(z, Y) = \sin \theta_0 + i\theta_0 z - \int_0^z (z - \zeta) \frac{i\pi}{2} H_0^{(1)}(\zeta, T) d\zeta, \quad (43)$$

with $\theta_0 = \arccos(1/Y)$; in general $(m+1)F_{m+2} - mF_m - izF_{m+1} + izF_{m-1} = e^{izY} Y^{-m-1} \sqrt{Y^2 - 1}$ and $F_{m-1} = -i\partial_z F_m$. Two further verified facts locate the function precisely. $G(z) = F_0(z, Y)$ satisfies an *inhomogeneous* Bessel equation,

$$G'' + \frac{G'}{z} + G = \frac{i\sqrt{Y^2 - 1}}{z} e^{izY}, \quad (44)$$

whose forcing is not a power of z , so G is not a Lommel function; and variation of parameters gives

$$G(z, Y) = T J_0(z) + \frac{i\pi\sqrt{Y^2 - 1}}{2} \left[Y_0(z) \int_0^z e^{isY} J_0(s) ds - J_0(z) \int_0^z e^{isY} Y_0(s) ds \right], \quad (45)$$

in which the two integrals are the *incomplete Lipschitz–Hankel integrals* Je_0 , Ye_0 of Dvorak and Dudley [4], for which published series algorithms exist [5]. Verified to 5×10^{-15} in double precision.

Other coplanar pairs. The flat-edge and coplanar-vertex pairs have trapezoidal convolution weights whose interior kinks are lines, not rays from the origin, so the sec-cancellation fails on the inner sub-rectangles and $\int \sec \theta e^{ikL \sec \theta}$ and $\int \sec^2 \theta e^{ikL \sec \theta}$ survive. These are F_0 and F_{-1} , i.e. incomplete Hankel functions of orders 0 and 1. The general coplanar case therefore needs F_{-1}, F_0, F_1, F_2 at several arguments: the same class, more instances.

Perpendicular pairs. Here the difference integral is genuinely three-dimensional. The radial part is still elementary, but the angular part is a two-dimensional integral over a spherical rectangle of $e^{ikR(\Omega)}$ times a rational function of the direction cosines. Doing one angle first yields an incomplete-Hankel-type function whose argument *and* endpoint both depend on the remaining angle, and no recursion collapses that. I found no reduction to one-variable functions, and the analogous problem in high-frequency diffraction theory (vertex diffraction) is likewise governed by a two-variable canonical function, the generalized Fresnel integral, with its own dedicated algorithms. My expectation, not a proof, is that the perpendicular pairs define a genuinely two-variable function of Appell or Kampé de Fériet type.

Which special functions are safe to leave in a result. A useful operational criterion is: does a rigorous arbitrary-precision implementation exist? Arb/FLINT (reachable from Julia through ArbLib.jl or ArbNumerics.jl) provides ball arithmetic with certified error bounds for Γ , the incomplete gamma and beta functions, the generalized exponential integral E_s , Ei/Si/Ci, erf and Fresnel, Bessel J, Y, I, K and hence the Hankel functions, Airy, ${}_0F_1$, ${}_1F_1$, U , ${}_2F_1$, general ${}_pF_q$, the elliptic integrals, and polylogarithms. Those are “chill”. The incomplete Hankel and Macdonald functions, the incomplete Lipschitz–Hankel integrals, and our F_m for $m \geq 1$ are absent from Arb, from mpmath, and from Mathematica’s Bessel-related function list, and I found no Julia package for them. They are “closed form” in the sense of Agrest and Maksimov, not in the sense of Arb. Can F_m be written through the safe list? Not as a finite combination: expanding in $x = 1 - 1/y^2$ gives $F_m = \frac{1}{2} \sum_n \frac{(iz)^n}{n!} B_X\left(\frac{1}{2}, \frac{m-n}{2}\right)$ with $X = \sin^2 \theta_0$, a series in z whose coefficients are incomplete beta functions of the other variable. Each coefficient is safe; the sum is a two-variable object, and summing it is numerically the same act, with the same cancellation, as the Taylor series in k .

Is the special-function form more robust than the series? Measured on the unit self panel ($D = \sqrt{2}$), Taylor series in Float64 with exact moments against the reduced form (37) in Float64 with 24-node Gauss–Legendre, both against a 300-bit reference:

kD	series terms	series digits lost	series error	reduced-form error
0.01	6	0	exact	5×10^{-11}
0.1	9	0	2×10^{-16}	8×10^{-13}
0.5	13	0	3×10^{-16}	1×10^{-14}
1	16	0.01	2×10^{-16}	4×10^{-16}
3	25	0.06	exact	2×10^{-16}
5	32	0.25	4×10^{-16}	2×10^{-16}
8	41	0.95	2×10^{-15}	7×10^{-17}
20	75	4.7	4×10^{-12}	exact
40	no convergence	8.5	garbage	1×10^{-16}

The two forms fail in mirror image. The series loses $\log_{10} \max_n |(kD)^n/n!|$ digits, negligible below $kD \approx 5$ and fatal by $kD = 40$. The reduced form carries an elementary block of size $4/k^3$ against an answer of size p^3 , so it loses about $3 \log_{10}(1/kD)$ digits as $kD \rightarrow 0$, and more quadrature nodes do not help because the loss is cancellation. Curing it means expanding the F ’s and E ’s in powers of k and cancelling analytically, which *is* the Taylor series. Gila’s design regime, cells from $\lambda/128$ to $\lambda/8$, is kD from 0.05 to about 1.1 across a face diagonal: exactly where the series is exact to the last bit with at most 16 terms and the special-function form loses one to three digits. The special-function form wins above $kD \approx 5$, i.e. cells above half a wavelength, where a piecewise-constant volume discretization is not credible anyway. Its genuine advantages are as a validation reference (it certified the four-dimensional evaluator to 10^{-60} here), and as the large- kD half of a hybrid if a single all- kD formula is ever wanted. The series also has a structural advantage the closed form cannot match: its moments do not depend on k , so a frequency sweep costs nothing.

Conclusion. For the coplanar pairs the exact answer is “elementary plus exponential integrals plus incomplete Hankel functions”, a statement that is true and precise but does not deliver arbitrary- k robustness, because the incomplete Hankel functions have no certified evaluator and their small-argument evaluation is the Taylor series in disguise. For the perpendicular pairs no one-variable special function form was found. The series of Section 5.1 is therefore not a fallback but the right representation for the regime Gila operates in, and the closed forms of the moments are its coefficients.

5.4 Feasibility of the series in floating point

A series is only useful if it converges in few terms and does not cancel. Both were measured (September 2026) with a 220-bit evaluator of the moments (Section 6.6). The number of terms N needed for 10^{-15} relative accuracy, and the digits lost to cancellation ($\log_{10}$ of the largest $|\text{term}|/|\text{sum}|$), at $f = 1 + 0.1i$, worst over face orientations:

cell	self	flat edge	cornered	coplanar vtx	perp. vtx	coplanar gap*	parallel offset
$\lambda/128$	7, 0.00	8, 0.00	7, 0.00	8, 0.00	8, 0.00	9, 0.00	9, 0.00
$\lambda/32$	9, 0.00	11, 0.01	10, 0.01	12, 0.01	12, 0.01	13, 0.02	13, 0.02
$\lambda/16$	12, 0.01	14, 0.02	13, 0.02	15, 0.03	15, 0.02	17, 0.04	16, 0.04
$\lambda/8$	15, 0.02	18, 0.05	16, 0.04	20, 0.09	19, 0.06	22, 0.28	20, 0.29
$\lambda/4$	19, 0.06	25, 0.30	22, 0.17	28, 0.58	26, 0.44	31, 0.97	28, 0.92
$\frac{1}{32} \times \frac{1}{32} \times \frac{1}{512}$	9, 0.00	11, 0.01	10, 0.01	12, 0.01	11, 0.01	13, 0.02	13, 0.02

*A coplanar pair with a one-width gap does not occur on the lattice (Section 3); it was measured as a representative near-singular pair. The laterally shifted parallel and non-meeting perpendicular pairs of diagonal neighbours were not measured; their diameters are no larger, so the same behaviour is expected.

Observations:

- N tracks $2\pi|f|D$ with D the diameter of the pair, as expected for e^{ikr} . Half the terms are even powers and free, so $\lambda/4$ needs about 13 odd moments per geometry and $\lambda/32$ needs 5 or 6.
- Cancellation is under one digit everywhere. The series is not alternating in the dangerous sense: its terms have magnitudes $(kD)^n/n!$ with a maximum of order $e^{kD}/\sqrt{2\pi kD}$ relative to the sum, and $kD \leq 2\pi\sqrt{3}/4 \approx 2.7$ at $\lambda/4$. For coarser cells the loss would grow like $kD/\ln 10$ digits, so the series should stay behind a cell-size gate with DIRECTFN as fallback.
- The slender cell is the *easiest* case: its diameters are small, so fewer terms and no cancellation. Slenderness hurts only the quadrature side, which the series removes.
- Complex frequency at 10% imaginary part adds at most one term and 0.4 digits. A large imaginary part increases $|f|$ and hence N but not the cancellation, since the envelope $e^{|k|r}$ dominates the sum rather than cancelling against it.

Agreement with Gila: the assembled series values were compared with **wekS**, **wekE**, **wekV** at **intOrd** 48 and 64. In every row the order-64 value lies between the order-48 value and the series, so Gila’s numerics converge toward the series value. This also gives the first honest error estimate of the current default: 6×10^{-11} for the self and edge integrals at $\lambda/4$ (9×10^{-13} at $\lambda/32$), 7×10^{-11} for the vertex integrals at all scales, and 1.8×10^{-9} for the vertex integrals on the slender cell. One flag: on the slender cell the flat-edge entry differs from the series by 3×10^{-13} while its own 48-to-64 spread is 1.3×10^{-14} , i.e. the DIRECTFN transformations plateau on sliver triangles and order refinement underestimates the error there by a factor 25. Refinement-based convergence tests are not a reliable error proxy for slender cells.

Cost: a **wekS+wekE+wekV** build at order 48 takes about 5 s per cubic cell shape and 20–35 s per slender shape on six threads. The series would be microseconds.

5.5 Separated pairs: why the series is the wrong tool there

For cells two or more apart, kD grows with the separation and the series (25) needs about 30 terms at $kD \approx 5$ while beginning to lose digits; the fixed 4–6 point Gauss–Legendre rules are cheaper and already reach the 10^{-10} target. The alternative expansion for far pairs is in the cell size at fixed centre separation $\mathbf{R}$: Taylor-expand $g(|\mathbf{R} + \boldsymbol{\delta}|)$ in $\boldsymbol{\delta}$, whose coefficients are derivatives of g at $\mathbf{R}$ times polynomial moments of the face pair, which are exact. This is the multipole idea, and it

converges when $|\delta| < |\mathbf{R}|$; at two cells the ratio is about 0.9 and convergence is slow, at four cells and beyond it would work. Since the fixed rule already costs little, I would not pursue it unless a uniform 10^{-14} far field becomes a goal.

6 Techniques for the remaining integrals

What follows is a toolbox, ordered roughly by how I would use it.

6.1 Difference variables and convolution weights

Section 4.1 in general form. For each axis, the pair of intervals $[a_1, a_2]$ (target) and $[b_1, b_2]$ (source) contributes a weight in the difference $u = x' - x$ equal to the convolution of their indicators, a trapezoid supported on $[b_1 - a_2, b_2 - a_1]$ with slopes ± 1 and plateau at height $\min(a_2 - a_1, b_2 - b_1)$. If one interval is a point (a rectangle in a perpendicular plane has zero extent along one axis) the weight is an indicator. The four-dimensional integral becomes a three-dimensional integral over the difference vector with a product of three such weights, of which at most two are trapezoids and at least one is an indicator or a point. Splitting each trapezoid at its breakpoints and reflecting into the positive octant yields a sum of integrals over boxes with multilinear weights, some of which have a corner at the origin (where $r = 0$) and the rest of which are regular. This decomposition was the basis of the numerical evaluator of Section 6.6 and is equally the basis of a symbolic derivation.

6.2 Polar and Duffy coordinates make the radial integral exact

On a two-dimensional box with a corner at the origin, polar coordinates give $r^m \rho \, d\rho \, d\theta$ with $r = \rho$, so the radial integral of a monomial is a power of the boundary radius $p \sec \theta$ or $q \csc \theta$ and the angular integrals are of $\sec^j \theta$ and $\csc^j \theta$, which satisfy the standard reduction $\int \sec^j = \frac{\sec^{j-2} \tan}{j-1} + \frac{j-2}{j-1} \int \sec^{j-2}$ and bottom out in $\int \sec = \operatorname{asinh} \tan$ and $\int \sec^2 = \tan$. In three dimensions, the Duffy transform $(x, y, z) = t(1, \eta, \zeta)$ on the pyramid with apex at the origin and base $x = p$ does the same job: $r = t\sqrt{1 + \eta^2 + \zeta^2}$ and the t -integral of t^{m+2} is exact, leaving a two-dimensional integral of a power of $\sqrt{1 + \eta^2 + \zeta^2}$ times a polynomial over a rectangle, which is again elementary by Section 6.2 applied once more. This is the route that produces the asinh and atan terms of N_0 .

For a recurrence in m , note that in polar coordinates the moment with kernel ρ^{2m+1} and weight $(p - \rho \cos \theta)(q - \rho \sin \theta)$ has radial integrals that are fixed polynomials in the boundary radius, so the moment is a linear combination of terms $p^i q^j h^l$ and of the two asinh terms with polynomial coefficients, with coefficients that follow a two-term recurrence in m from the $\sec^j$ reduction. Asking Mathematica for the moments at $m = 0, \dots, 4$ individually and pattern-matching the coefficients is the fastest way to obtain the recurrence; asking for symbolic m produces hypergeometric ${}_2F_1$ s with half-integer parameters and unusable conditions, as was observed.

6.3 The divergence recursion as an algorithm

Theorem 5.1 is constructive and can be implemented as a small symbolic program, which is the most systematic way to get all eight geometries and all odd m at once:

1. Represent the domain as the product box $F \times F'$ in the four free coordinates, and the kernel as $(|AX|^2 + c^2)^{m/2}$ with A the difference map (initially $c = 0$).
2. Apply (27): replace the integral by $\frac{1}{N+m} \sum_Q (X_Q \cdot \nu_Q) \int_Q$ plus, when $c \neq 0$, $-\frac{mc^2}{N+m} \int (\dots)^{(m-2)/2}$ over the same domain.

3. On each facet, restrict A , drop null directions (they contribute a length factor), complete the square to find the new c , and recurse.
4. At dimension one, integrate $(\beta^2 t^2 + c^2)^{m'/2}$ in closed form.

The output is a sum of products of lengths with asinh and atan of ratios of lengths and of square roots of sums of squares of lengths, i.e. exactly the shape of (15)–(23). A computer-algebra implementation should take a day; its advantage over asking Mathematica for each integral is that the structure of the result is controlled and the cancellation-free regrouping (next subsection) can be applied mechanically. For the terminal one-dimensional integrals, note that $\int_0^L (t^2 + c^2)^{m'/2} dt$ for odd m' obeys the recurrence

$$J_{m'} = \frac{L(L^2 + c^2)^{m'/2} + m'c^2 J_{m'-2}}{m' + 1}, \quad J_{-1} = \operatorname{asinh}(L/c), \quad (46)$$

which follows from the same divergence identity in one dimension.

6.4 Integration by parts against a known potential

When one panel's coordinates cannot be paired with the other's (perpendicular planes), reduce first to the potential of the source rectangle at a point, $\Phi_m(\mathbf{x}) = \int_{F'} |\mathbf{x} - \mathbf{y}|^m dS'$, then integrate Φ_m over the target rectangle. For the shared-edge direction the weight $(l_a - u)$ can be removed by parts, as in Section 4.4, at the cost of one more antiderivative in u of an already elementary function. The classical box potentials N_0 (22) and $W(a, b, c) = \int_0^b \int_0^c \sqrt{a^2 + y^2 + z^2}$ (the $m = 1$ analogue, appearing in the derivation of (23)) are the first two members of a family $\int_{\text{box}} r^m$ that the divergence recursion generates.

6.5 Cancellation-free regrouping

Every closed form obtained by the above will, as written, cancel catastrophically for slender panels: the individual terms scale like (largest length) $^{m+4}$ while the integral over a slender pair scales with the small lengths. The measured losses of 7–19 digits in the original Gila formulas are typical. The regrouping rules that fixed them, in order of importance:

1. **Never subtract a length from a hypotenuse.** $\sqrt{x^2 + y^2} - x = y^2/(\sqrt{x^2 + y^2} + x)$, and $h^3 - q^3 = p^2(h^2 + hq + q^2)/(h + q)$. Apply recursively to any polynomial in h, p, q that vanishes as $p \rightarrow 0$.
2. **Prefer asinh to differences of logarithms**, and write differences of asinh as $\log(1 + x)$ with x a ratio of positive quantities obtained by rationalizing. $\operatorname{asinh}(a) - \operatorname{asinh}(b) = \log \frac{a + \sqrt{1 + a^2}}{b + \sqrt{1 + b^2}}$, and the ratio minus one can be rationalized to a manifestly positive expression.
3. **Keep atan arguments as ratios of positive quantities**, never as $\operatorname{atan}(1/x)$ or acot of something that changes sign.
4. **Find the exact identities first.** In the self integral every $\sqrt{l_a^2 + 4l_b^2}$ cancelled exactly; in the cornered integral the l_b^3 asinh terms cancelled against the leading part of the $\operatorname{asinh}(l_c/h_{ab})$ term. The diagnostic is the amplification $\sum |\text{term}_i| / |\sum \text{term}_i|$ in extended precision, evaluated over the aspect-ratio grid: if it explains the digits lost, the loss is between top-level terms and the fix is to group them; if the top-level amplification is one but digits are still lost, the loss is inside a sub-expression and rule 1 or 2 applies to it.
5. **Make symmetric integrals symmetric in form.** Mathematica's output is usually asymmetric in symmetric arguments, and the asymmetry is where the cancellation hides.
6. **When all else fails, extended precision.** A BigFloat evaluation of a badly conditioned formula rounded to Float64 costs microseconds per call and is perfectly acceptable for a

handful of constants per cell shape, but it leaves a fragile expression in the code and needs the typed-constant care of Section 4.7.

6.6 A high-precision evaluator for validation

Whatever closed forms are derived must be checked against something independent, and the tool that worked is a numerical evaluator of all the moments I_m of a rectangle pair, in 220-bit arithmetic. Its construction: reduce to the difference vector with the convolution weights of Section 4.1; split at the breakpoints and reflect into the positive octant to get boxes with multilinear weights; for a box with a corner at the origin apply the Duffy transform, under which the radial integrand is an exact polynomial in the Duffy variable for every integer $m \geq -1$, and integrate the remaining smooth two-dimensional integrand with a tensor Gauss–Legendre rule; for boxes away from the origin use a graded tensor rule (grading toward the near corner is essential for slender cells, which lost twelve digits without it). It was validated against the exact even moments to 10^{-58} and against the $1/r$ closed forms to 10^{-30} or better, and its series sum agreed with a direct high-precision integration of the full kernel to 10^{-30} . The same code can serve as the reference for the tests of any new closed form, and its structure is also the skeleton of the symbolic derivation.

6.7 Using Mathematica effectively

- Integrate fixed integer m one at a time with `Assumptions -> {la > 0, lb > 0, lc > 0}`. Symbolic m yields ${}_2F_1$ s.
- Do not expect `FullSimplify` to prove two elementary forms equal when they differ by logarithm identities such as $(h - l_a)(h + l_a) = l_b^2$; it typically leaves a residue. Prove equality numerically at 100 digits on a grid of extreme aspect ratios instead (the residues we saw were 10^{-139}), or help it with `TrigToExp` and `PowerExpand` under the positivity assumptions.
- The iterated one-dimensional route (integrate in t , then in the next variable, with the potential functions N_0 , W substituted by hand) gives cleaner output than a direct multidimensional `Integrate`.
- Get the moment at several m and fit the recurrence; do not ask for the general term. Section 5.2 shows what the fit looks like for the self face, and that Mathematica’s low- m outputs must be rewritten before use: their logarithm groupings are exactly equal to single small `asinh` values and lose all significance for slender panels.

6.8 An implementation sketch for Gila

If the odd moments are derived, the natural integration point is `wekTrp`, which already memoizes the weak integrals per cell shape. The series replacement would:

1. compute, once per cell shape and independently of frequency, the moments I_m for $m = -1, 0, 1, \dots, M$ for each of the eight geometries and three face orientations (a few hundred real numbers);
2. for a given f , sum $\frac{1}{4\pi f^2} \sum_n \frac{(2\pi i f)^n}{n!} I_{n-1}$ to convergence, in `Float64`, with a gate on $2\pi|f|D$ that falls back to `DIRECTFN` above about $kD \approx 3$;
3. assemble the `wS`, `wE`, `wV` vectors with the existing conventions (raw integrals divided by cell volume; edge and vertex cross terms averaged over the two orientations, which become exactly equal);
4. replace the order-9 rule for the touching-shell pairs with the same series, after measuring the two unmeasured geometries.

The frequency independence of the moments is a real bonus: a frequency sweep on a fixed geometry would rebuild the contact terms at no cost. Tests should compare each new closed form with the extended-precision evaluator on an aspect-ratio grid, check homogeneity in f , and check the assembled operator against the current DIRECTFN values at two orders, which must bracket the new value.

7 Collected numerical facts

For reference, the measurements that shaped the current code.

- **Digits lost by the original $1/r$ closed forms**, by aspect ratio l_a/l_b : self 1.9 at $1/32$, 4.5 at 10^{-3} , 6.8 at 10^{-4} ; flat edge 4.0 at 10^{-1} , 8.1 at 10^{-2} , 11.9 at 10^{-3} , 15.2 at 10^{-4} (i.e. no correct digits); cornered edge up to 18.8 at $l_a \ll l_b \ll l_c$. The new forms: ≤ 0.2 , ≤ 0.2 , ≤ 0.5 over ratios 10^{-8} to 10^8 .
- **Fixed Gauss–Legendre orders** needed for 10^{-10} assembled error at separations 2, 3, 4, 5, 6, 8–16, ≥ 24 cells: 8, 6, 6, 5, 5, 4, 3, with a cell-size floor of 3, 4, 5 at $\leq \lambda/32$, $\lambda/8$, $\lambda/4$. Convergence with order is about 0.4–0.8 decades per point, roughly half the Bernstein-ellipse prediction. Fixed rules are about 1000 times cheaper than adaptive cubature at the same accuracy for separations ≥ 2 .
- **Assembly amplification** of face-pair error into tensor error: 10–1000.
- **DIRECTFN at int0rd 48**: true relative error 6×10^{-11} (self/edge, $\lambda/4$), 9×10^{-13} (self/edge, $\lambda/32$), 7×10^{-11} (vertex, all scales), 1.8×10^{-9} (vertex, slender 16:1 cell). Cost 5 s per cubic cell shape, 20–35 s per slender shape.
- **Regularized kernel** $\text{cispi}(x) \text{sinc}(x)/(2f)$: about 2ϵ against a 256-bit reference for real and complex f . Far kernel with `cispi`: 0.07 ulp at $r = 100$, $f = 1$, against 29 ulp for `cis`.
- **Series feasibility**: Section 5.4.
- **Anti-Hermitian part**: Section 2.4.

8 Open questions and risks

- **Coarse cells**. Above $\lambda/4$ the series loses about $kD/\ln 10$ digits; the gate and the DIRECTFN fallback must stay. Whether anyone uses cells that coarse with a contact treatment is a separate question.
- **Sliver triangles in DIRECTFN**. The plateau on slender cells means the current vertex integrals are only good to 10^{-9} there, and the convergence test in `intConTest.jl` cannot see it. The series fixes this; until then, slender cells carry that error.
- **Vertex conventions**. The vertex geometries were reconstructed from `wekVDir` and `wekGrdPts!` and the series matched `wekV` to the reference accuracy, so the reconstruction is right, but the entry-to-geometry map should be re-derived and documented when the closed forms are implemented.
- **Complex frequency with large imaginary part**. Only $|\text{Im}f| = 0.1$ was measured. The theory says $N \propto |f|D$ with benign cancellation, but it should be checked at the largest imaginary parts in use before the DIRECTFN fallback is removed.
- **The fixed-rule schedule was measured at real frequency only**. With complex f the far kernel gains a real exponential factor; the schedule uses $|f|$ in its cell-size argument, which is a reasonable proxy but unmeasured.
- **The perpendicular full-kernel integral**. Whether it reduces to a known two-variable special function is open; nothing in Gila needs the answer, since the series covers it, but it

would settle the classification.

- **The repository environment.** At the time of writing the checked-out `Manifest.toml` was stale relative to the Metal-backend `Project.toml` and the package failed to precompile from the repo; the studies ran from throwaway environments that `dev` the repo. A `Pkg.resolve()` fixes it.
- **A floating-point gate in the contact path.** `egoCntOut!` selects sub-cell windows by rounding a `Float64` ratio of a separation to a very fine gcd cell scale; the inputs are exact rationals upstream and the arithmetic could be exact. Not an accuracy issue but a wrong-branch risk.

Acknowledgement of method

The derivations of Section 4, the general-moment analysis of Section 5.2, the special-function reduction of Section 5.3, and the measurement studies were carried out with the help of subagents running Claude Opus, and independently checked by me in extended precision; the symbolic equivalences were additionally checked by Paul Virally in Mathematica to 100 digits. The theorem of Section 5.1 is my synthesis of the classical polyhedral-potential argument for the rectangle-pair setting and has been checked only against the cases $m = -1$ (three geometries) and $m = 1$ (cornered geometry) that were derived explicitly, and numerically for all m through the evaluator; it should be treated as a well-founded plan rather than a finished proof until the first odd moments have been written out.

A The 36 face-pair integrals, written out

Source cell centred at the origin with edges s_1, s_2, s_3 ; target cell centred at $(\Delta_1 s_1, \Delta_2 s_2, \Delta_3 s_3)$ with the same edges (integer Δ for a regular grid, but the formulas hold for any offset). Target-face coordinates are ξ , source-face coordinates η , both measured from the respective cell centre, so every limit is $\pm s_d/2$ and the offset appears only inside the kernel. The face that is fixed in a coordinate contributes the constant $\pm s_d/2$ for that coordinate. g is the kernel (1). The nine tensor components

are the signed sums

$$\begin{aligned}
(\mathbf{G}_0)_{xx} &= \frac{1}{k^2 V_t} \left[I_{xzL,xzL} - I_{xzL,xzU} - I_{xzU,xzL} + I_{xzU,xzU} \right. \\
&\quad \left. + I_{xyL,xyL} - I_{xyL,xyU} - I_{xyU,xyL} + I_{xyU,xyU} \right] - \frac{\delta_{ts}}{k^2} \\
(\mathbf{G}_0)_{xy} &= \frac{1}{k^2 V_t} \left[-I_{yzL,xzL} + I_{yzL,xzU} + I_{yzU,xzL} - I_{yzU,xzU} \right] \\
(\mathbf{G}_0)_{xz} &= \frac{1}{k^2 V_t} \left[-I_{yzL,xyL} + I_{yzL,xyU} + I_{yzU,xyL} - I_{yzU,xyU} \right] \\
(\mathbf{G}_0)_{yx} &= \frac{1}{k^2 V_t} \left[-I_{xzL,yzL} + I_{xzL,yzU} + I_{xzU,yzL} - I_{xzU,yzU} \right] \\
(\mathbf{G}_0)_{yy} &= \frac{1}{k^2 V_t} \left[I_{yzL,yzL} - I_{yzL,yzU} - I_{yzU,yzL} + I_{yzU,yzU} \right. \\
&\quad \left. + I_{xyL,xyL} - I_{xyL,xyU} - I_{xyU,xyL} + I_{xyU,xyU} \right] - \frac{\delta_{ts}}{k^2} \\
(\mathbf{G}_0)_{yz} &= \frac{1}{k^2 V_t} \left[-I_{xzL,xyL} + I_{xzL,xyU} + I_{xzU,xyL} - I_{xzU,xyU} \right] \\
(\mathbf{G}_0)_{zx} &= \frac{1}{k^2 V_t} \left[-I_{xyL,yzL} + I_{xyL,yzU} + I_{xyU,yzL} - I_{xyU,yzU} \right] \\
(\mathbf{G}_0)_{zy} &= \frac{1}{k^2 V_t} \left[-I_{xyL,xzL} + I_{xyL,xzU} + I_{xyU,xzL} - I_{xyU,xzU} \right] \\
(\mathbf{G}_0)_{zz} &= \frac{1}{k^2 V_t} \left[I_{yzL,yzL} - I_{yzL,yzU} - I_{yzU,yzL} + I_{yzU,yzU} \right. \\
&\quad \left. + I_{xzL,xzL} - I_{xzL,xzU} - I_{xzU,xzL} + I_{xzU,xzU} \right] - \frac{\delta_{ts}}{k^2}
\end{aligned}$$

with $V_t = s_1 s_2 s_3$ and the delta term present only for $\Delta = 0$. The 36 integrals follow, grouped by target face; Table 1 says which geometry each one has for a given Δ .

Target face yzL (normal $-\mathbf{e}_x$)

$$\begin{aligned}
I_{yzL,yzL} &= \int_{-s_2/2}^{s_2/2} \int_{-s_3/2}^{s_3/2} \int_{-s_2/2}^{s_2/2} \int_{-s_3/2}^{s_3/2} g\left(\sqrt{(\Delta_1 s_1)^2 + (\Delta_2 s_2 + \xi_2 - \eta_2)^2 + (\Delta_3 s_3 + \xi_3 - \eta_3)^2}\right) d\eta_3 d\eta_2 d\xi_3 d\xi_2 \\
I_{yzL,yzU} &= \int_{-s_2/2}^{s_2/2} \int_{-s_3/2}^{s_3/2} \int_{-s_2/2}^{s_2/2} \int_{-s_3/2}^{s_3/2} g\left(\sqrt{(\Delta_1 s_1 - s_1)^2 + (\Delta_2 s_2 + \xi_2 - \eta_2)^2 + (\Delta_3 s_3 + \xi_3 - \eta_3)^2}\right) d\eta_3 d\eta_2 d\xi_3 d\xi_2 \\
I_{yzL,xzL} &= \int_{-s_2/2}^{s_2/2} \int_{-s_3/2}^{s_3/2} \int_{-s_1/2}^{s_1/2} \int_{-s_3/2}^{s_3/2} g\left(\sqrt{(\Delta_1 s_1 - \frac{s_1}{2} - \eta_1)^2 + (\Delta_2 s_2 + \xi_2 + \frac{s_2}{2})^2 + (\Delta_3 s_3 + \xi_3 - \eta_3)^2}\right) d\eta_3 d\eta_1 d\xi_3 d\xi_2 \\
I_{yzL,xzU} &= \int_{-s_2/2}^{s_2/2} \int_{-s_3/2}^{s_3/2} \int_{-s_1/2}^{s_1/2} \int_{-s_3/2}^{s_3/2} g\left(\sqrt{(\Delta_1 s_1 - \frac{s_1}{2} - \eta_1)^2 + (\Delta_2 s_2 + \xi_2 - \frac{s_2}{2})^2 + (\Delta_3 s_3 + \xi_3 - \eta_3)^2}\right) d\eta_3 d\eta_1 d\xi_3 d\xi_2 \\
I_{yzL,xyL} &= \int_{-s_2/2}^{s_2/2} \int_{-s_3/2}^{s_3/2} \int_{-s_1/2}^{s_1/2} \int_{-s_2/2}^{s_2/2} g\left(\sqrt{(\Delta_1 s_1 - \frac{s_1}{2} - \eta_1)^2 + (\Delta_2 s_2 + \xi_2 - \eta_2)^2 + (\Delta_3 s_3 + \xi_3 + \frac{s_3}{2})^2}\right) d\eta_2 d\eta_1 d\xi_3 d\xi_2 \\
I_{yzL,xyU} &= \int_{-s_2/2}^{s_2/2} \int_{-s_3/2}^{s_3/2} \int_{-s_1/2}^{s_1/2} \int_{-s_2/2}^{s_2/2} g\left(\sqrt{(\Delta_1 s_1 - \frac{s_1}{2} - \eta_1)^2 + (\Delta_2 s_2 + \xi_2 - \eta_2)^2 + (\Delta_3 s_3 + \xi_3 - \frac{s_3}{2})^2}\right) d\eta_2 d\eta_1 d\xi_3 d\xi_2
\end{aligned}$$

Target face yzU (normal $+\mathbf{e}_x$)

$$\begin{aligned}
I_{yzU,yzL} &= \int_{-s_2/2}^{s_2/2} \int_{-s_3/2}^{s_3/2} \int_{-s_2/2}^{s_2/2} \int_{-s_3/2}^{s_3/2} g\left(\sqrt{(\Delta_1 s_1 + s_1)^2 + (\Delta_2 s_2 + \xi_2 - \eta_2)^2 + (\Delta_3 s_3 + \xi_3 - \eta_3)^2}\right) d\eta_3 d\eta_2 d\xi_3 d\xi_2 \\
I_{yzU,yzU} &= \int_{-s_2/2}^{s_2/2} \int_{-s_3/2}^{s_3/2} \int_{-s_2/2}^{s_2/2} \int_{-s_3/2}^{s_3/2} g\left(\sqrt{(\Delta_1 s_1)^2 + (\Delta_2 s_2 + \xi_2 - \eta_2)^2 + (\Delta_3 s_3 + \xi_3 - \eta_3)^2}\right) d\eta_3 d\eta_2 d\xi_3 d\xi_2 \\
I_{yzU,xzL} &= \int_{-s_2/2}^{s_2/2} \int_{-s_3/2}^{s_3/2} \int_{-s_1/2}^{s_1/2} \int_{-s_3/2}^{s_3/2} g\left(\sqrt{(\Delta_1 s_1 + \frac{s_1}{2} - \eta_1)^2 + (\Delta_2 s_2 + \xi_2 + \frac{s_2}{2})^2 + (\Delta_3 s_3 + \xi_3 - \eta_3)^2}\right) d\eta_3 d\eta_1 d\xi_3 d\xi_2 \\
I_{yzU,xzU} &= \int_{-s_2/2}^{s_2/2} \int_{-s_3/2}^{s_3/2} \int_{-s_1/2}^{s_1/2} \int_{-s_3/2}^{s_3/2} g\left(\sqrt{(\Delta_1 s_1 + \frac{s_1}{2} - \eta_1)^2 + (\Delta_2 s_2 + \xi_2 - \frac{s_2}{2})^2 + (\Delta_3 s_3 + \xi_3 - \eta_3)^2}\right) d\eta_3 d\eta_1 d\xi_3 d\xi_2 \\
I_{yzU,xyL} &= \int_{-s_2/2}^{s_2/2} \int_{-s_3/2}^{s_3/2} \int_{-s_1/2}^{s_1/2} \int_{-s_2/2}^{s_2/2} g\left(\sqrt{(\Delta_1 s_1 + \frac{s_1}{2} - \eta_1)^2 + (\Delta_2 s_2 + \xi_2 - \eta_2)^2 + (\Delta_3 s_3 + \xi_3 + \frac{s_3}{2})^2}\right) d\eta_2 d\eta_1 d\xi_3 d\xi_2 \\
I_{yzU,xyU} &= \int_{-s_2/2}^{s_2/2} \int_{-s_3/2}^{s_3/2} \int_{-s_1/2}^{s_1/2} \int_{-s_2/2}^{s_2/2} g\left(\sqrt{(\Delta_1 s_1 + \frac{s_1}{2} - \eta_1)^2 + (\Delta_2 s_2 + \xi_2 - \eta_2)^2 + (\Delta_3 s_3 + \xi_3 - \frac{s_3}{2})^2}\right) d\eta_2 d\eta_1 d\xi_3 d\xi_2
\end{aligned}$$

Target face xzL (normal $-\mathbf{e}_y$)

$$\begin{aligned}
I_{xzL,yzL} &= \int_{-s_1/2}^{s_1/2} \int_{-s_3/2}^{s_3/2} \int_{-s_2/2}^{s_2/2} \int_{-s_3/2}^{s_3/2} g\left(\sqrt{(\Delta_1 s_1 + \xi_1 + \frac{s_1}{2})^2 + (\Delta_2 s_2 - \frac{s_2}{2} - \eta_2)^2 + (\Delta_3 s_3 + \xi_3 - \eta_3)^2}\right) d\eta_3 d\eta_2 d\xi_3 d\xi_1 \\
I_{xzL,yzU} &= \int_{-s_1/2}^{s_1/2} \int_{-s_3/2}^{s_3/2} \int_{-s_2/2}^{s_2/2} \int_{-s_3/2}^{s_3/2} g\left(\sqrt{(\Delta_1 s_1 + \xi_1 - \frac{s_1}{2})^2 + (\Delta_2 s_2 - \frac{s_2}{2} - \eta_2)^2 + (\Delta_3 s_3 + \xi_3 - \eta_3)^2}\right) d\eta_3 d\eta_2 d\xi_3 d\xi_1 \\
I_{xzL,xzL} &= \int_{-s_1/2}^{s_1/2} \int_{-s_3/2}^{s_3/2} \int_{-s_1/2}^{s_1/2} \int_{-s_3/2}^{s_3/2} g\left(\sqrt{(\Delta_1 s_1 + \xi_1 - \eta_1)^2 + (\Delta_2 s_2)^2 + (\Delta_3 s_3 + \xi_3 - \eta_3)^2}\right) d\eta_3 d\eta_1 d\xi_3 d\xi_1 \\
I_{xzL,xzU} &= \int_{-s_1/2}^{s_1/2} \int_{-s_3/2}^{s_3/2} \int_{-s_1/2}^{s_1/2} \int_{-s_3/2}^{s_3/2} g\left(\sqrt{(\Delta_1 s_1 + \xi_1 - \eta_1)^2 + (\Delta_2 s_2 - s_2)^2 + (\Delta_3 s_3 + \xi_3 - \eta_3)^2}\right) d\eta_3 d\eta_1 d\xi_3 d\xi_1 \\
I_{xzL,xyL} &= \int_{-s_1/2}^{s_1/2} \int_{-s_3/2}^{s_3/2} \int_{-s_1/2}^{s_1/2} \int_{-s_2/2}^{s_2/2} g\left(\sqrt{(\Delta_1 s_1 + \xi_1 - \eta_1)^2 + (\Delta_2 s_2 - \frac{s_2}{2} - \eta_2)^2 + (\Delta_3 s_3 + \xi_3 + \frac{s_3}{2})^2}\right) d\eta_2 d\eta_1 d\xi_3 d\xi_1 \\
I_{xzL,xyU} &= \int_{-s_1/2}^{s_1/2} \int_{-s_3/2}^{s_3/2} \int_{-s_1/2}^{s_1/2} \int_{-s_2/2}^{s_2/2} g\left(\sqrt{(\Delta_1 s_1 + \xi_1 - \eta_1)^2 + (\Delta_2 s_2 - \frac{s_2}{2} - \eta_2)^2 + (\Delta_3 s_3 + \xi_3 - \frac{s_3}{2})^2}\right) d\eta_2 d\eta_1 d\xi_3 d\xi_1
\end{aligned}$$

Target face xzU (normal $+\mathbf{e}_y$)

$$\begin{aligned}
I_{xzU,yzL} &= \int_{-s_1/2}^{s_1/2} \int_{-s_3/2}^{s_3/2} \int_{-s_2/2}^{s_2/2} \int_{-s_3/2}^{s_3/2} g\left(\sqrt{(\Delta_1 s_1 + \xi_1 + \frac{s_1}{2})^2 + (\Delta_2 s_2 + \frac{s_2}{2} - \eta_2)^2 + (\Delta_3 s_3 + \xi_3 - \eta_3)^2}\right) d\eta_3 d\eta_2 d\xi_3 d\xi_1 \\
I_{xzU,yzU} &= \int_{-s_1/2}^{s_1/2} \int_{-s_3/2}^{s_3/2} \int_{-s_2/2}^{s_2/2} \int_{-s_3/2}^{s_3/2} g\left(\sqrt{(\Delta_1 s_1 + \xi_1 - \frac{s_1}{2})^2 + (\Delta_2 s_2 + \frac{s_2}{2} - \eta_2)^2 + (\Delta_3 s_3 + \xi_3 - \eta_3)^2}\right) d\eta_3 d\eta_2 d\xi_3 d\xi_1 \\
I_{xzU,xzL} &= \int_{-s_1/2}^{s_1/2} \int_{-s_3/2}^{s_3/2} \int_{-s_1/2}^{s_1/2} \int_{-s_3/2}^{s_3/2} g\left(\sqrt{(\Delta_1 s_1 + \xi_1 - \eta_1)^2 + (\Delta_2 s_2 + s_2)^2 + (\Delta_3 s_3 + \xi_3 - \eta_3)^2}\right) d\eta_3 d\eta_1 d\xi_3 d\xi_1 \\
I_{xzU,xzU} &= \int_{-s_1/2}^{s_1/2} \int_{-s_3/2}^{s_3/2} \int_{-s_1/2}^{s_1/2} \int_{-s_3/2}^{s_3/2} g\left(\sqrt{(\Delta_1 s_1 + \xi_1 - \eta_1)^2 + (\Delta_2 s_2)^2 + (\Delta_3 s_3 + \xi_3 - \eta_3)^2}\right) d\eta_3 d\eta_1 d\xi_3 d\xi_1 \\
I_{xzU,xyL} &= \int_{-s_1/2}^{s_1/2} \int_{-s_3/2}^{s_3/2} \int_{-s_1/2}^{s_1/2} \int_{-s_2/2}^{s_2/2} g\left(\sqrt{(\Delta_1 s_1 + \xi_1 - \eta_1)^2 + (\Delta_2 s_2 + \frac{s_2}{2} - \eta_2)^2 + (\Delta_3 s_3 + \xi_3 + \frac{s_3}{2})^2}\right) d\eta_2 d\eta_1 d\xi_3 d\xi_1 \\
I_{xzU,xyU} &= \int_{-s_1/2}^{s_1/2} \int_{-s_3/2}^{s_3/2} \int_{-s_1/2}^{s_1/2} \int_{-s_2/2}^{s_2/2} g\left(\sqrt{(\Delta_1 s_1 + \xi_1 - \eta_1)^2 + (\Delta_2 s_2 + \frac{s_2}{2} - \eta_2)^2 + (\Delta_3 s_3 + \xi_3 - \frac{s_3}{2})^2}\right) d\eta_2 d\eta_1 d\xi_3 d\xi_1
\end{aligned}$$

Target face xyL (normal $-\mathbf{e}_z$)

$$\begin{aligned}
I_{xyL,yzL} &= \int_{-s_1/2}^{s_1/2} \int_{-s_2/2}^{s_2/2} \int_{-s_2/2}^{s_2/2} \int_{-s_3/2}^{s_3/2} g \left(\sqrt{(\Delta_1 s_1 + \xi_1 + \frac{s_1}{2})^2 + (\Delta_2 s_2 + \xi_2 - \eta_2)^2 + (\Delta_3 s_3 - \frac{s_3}{2} - \eta_3)^2} \right) d\eta_3 d\eta_2 d\xi_2 d\xi_1 \\
I_{xyL,yzU} &= \int_{-s_1/2}^{s_1/2} \int_{-s_2/2}^{s_2/2} \int_{-s_2/2}^{s_2/2} \int_{-s_3/2}^{s_3/2} g \left(\sqrt{(\Delta_1 s_1 + \xi_1 - \frac{s_1}{2})^2 + (\Delta_2 s_2 + \xi_2 - \eta_2)^2 + (\Delta_3 s_3 - \frac{s_3}{2} - \eta_3)^2} \right) d\eta_3 d\eta_2 d\xi_2 d\xi_1 \\
I_{xyL,xzL} &= \int_{-s_1/2}^{s_1/2} \int_{-s_2/2}^{s_2/2} \int_{-s_1/2}^{s_1/2} \int_{-s_3/2}^{s_3/2} g \left(\sqrt{(\Delta_1 s_1 + \xi_1 - \eta_1)^2 + (\Delta_2 s_2 + \xi_2 + \frac{s_2}{2})^2 + (\Delta_3 s_3 - \frac{s_3}{2} - \eta_3)^2} \right) d\eta_3 d\eta_1 d\xi_2 d\xi_1 \\
I_{xyL,xzU} &= \int_{-s_1/2}^{s_1/2} \int_{-s_2/2}^{s_2/2} \int_{-s_1/2}^{s_1/2} \int_{-s_3/2}^{s_3/2} g \left(\sqrt{(\Delta_1 s_1 + \xi_1 - \eta_1)^2 + (\Delta_2 s_2 + \xi_2 - \frac{s_2}{2})^2 + (\Delta_3 s_3 - \frac{s_3}{2} - \eta_3)^2} \right) d\eta_3 d\eta_1 d\xi_2 d\xi_1 \\
I_{xyL,xyL} &= \int_{-s_1/2}^{s_1/2} \int_{-s_2/2}^{s_2/2} \int_{-s_1/2}^{s_1/2} \int_{-s_2/2}^{s_2/2} g \left(\sqrt{(\Delta_1 s_1 + \xi_1 - \eta_1)^2 + (\Delta_2 s_2 + \xi_2 - \eta_2)^2 + (\Delta_3 s_3)^2} \right) d\eta_2 d\eta_1 d\xi_2 d\xi_1 \\
I_{xyL,xyU} &= \int_{-s_1/2}^{s_1/2} \int_{-s_2/2}^{s_2/2} \int_{-s_1/2}^{s_1/2} \int_{-s_2/2}^{s_2/2} g \left(\sqrt{(\Delta_1 s_1 + \xi_1 - \eta_1)^2 + (\Delta_2 s_2 + \xi_2 - \eta_2)^2 + (\Delta_3 s_3 - s_3)^2} \right) d\eta_2 d\eta_1 d\xi_2 d\xi_1
\end{aligned}$$

Target face xyU (normal $+\mathbf{e}_z$)

$$\begin{aligned}
I_{xyU,yzL} &= \int_{-s_1/2}^{s_1/2} \int_{-s_2/2}^{s_2/2} \int_{-s_2/2}^{s_2/2} \int_{-s_3/2}^{s_3/2} g \left(\sqrt{(\Delta_1 s_1 + \xi_1 + \frac{s_1}{2})^2 + (\Delta_2 s_2 + \xi_2 - \eta_2)^2 + (\Delta_3 s_3 + \frac{s_3}{2} - \eta_3)^2} \right) d\eta_3 d\eta_2 d\xi_2 d\xi_1 \\
I_{xyU,yzU} &= \int_{-s_1/2}^{s_1/2} \int_{-s_2/2}^{s_2/2} \int_{-s_2/2}^{s_2/2} \int_{-s_3/2}^{s_3/2} g \left(\sqrt{(\Delta_1 s_1 + \xi_1 - \frac{s_1}{2})^2 + (\Delta_2 s_2 + \xi_2 - \eta_2)^2 + (\Delta_3 s_3 + \frac{s_3}{2} - \eta_3)^2} \right) d\eta_3 d\eta_2 d\xi_2 d\xi_1 \\
I_{xyU,xzL} &= \int_{-s_1/2}^{s_1/2} \int_{-s_2/2}^{s_2/2} \int_{-s_1/2}^{s_1/2} \int_{-s_3/2}^{s_3/2} g \left(\sqrt{(\Delta_1 s_1 + \xi_1 - \eta_1)^2 + (\Delta_2 s_2 + \xi_2 + \frac{s_2}{2})^2 + (\Delta_3 s_3 + \frac{s_3}{2} - \eta_3)^2} \right) d\eta_3 d\eta_1 d\xi_2 d\xi_1 \\
I_{xyU,xzU} &= \int_{-s_1/2}^{s_1/2} \int_{-s_2/2}^{s_2/2} \int_{-s_1/2}^{s_1/2} \int_{-s_3/2}^{s_3/2} g \left(\sqrt{(\Delta_1 s_1 + \xi_1 - \eta_1)^2 + (\Delta_2 s_2 + \xi_2 - \frac{s_2}{2})^2 + (\Delta_3 s_3 + \frac{s_3}{2} - \eta_3)^2} \right) d\eta_3 d\eta_1 d\xi_2 d\xi_1 \\
I_{xyU,xyL} &= \int_{-s_1/2}^{s_1/2} \int_{-s_2/2}^{s_2/2} \int_{-s_1/2}^{s_1/2} \int_{-s_2/2}^{s_2/2} g \left(\sqrt{(\Delta_1 s_1 + \xi_1 - \eta_1)^2 + (\Delta_2 s_2 + \xi_2 - \eta_2)^2 + (\Delta_3 s_3 + s_3)^2} \right) d\eta_2 d\eta_1 d\xi_2 d\xi_1 \\
I_{xyU,xyU} &= \int_{-s_1/2}^{s_1/2} \int_{-s_2/2}^{s_2/2} \int_{-s_1/2}^{s_1/2} \int_{-s_2/2}^{s_2/2} g \left(\sqrt{(\Delta_1 s_1 + \xi_1 - \eta_1)^2 + (\Delta_2 s_2 + \xi_2 - \eta_2)^2 + (\Delta_3 s_3)^2} \right) d\eta_2 d\eta_1 d\xi_2 d\xi_1
\end{aligned}$$

References

- [1] A. G. Polimeridis, F. Vipiana, J. R. Mosig, D. R. Wilton, “DIRECTFN: Fully numerical algorithms for high precision computation of singular integrals in Galerkin SIE methods,” *IEEE Trans. Antennas Propag.* 61(6), 3112–3122 (2013).
- [2] J. Waldvogel, “The Newtonian potential of homogeneous polyhedra,” *Z. Angew. Math. Phys.* 30, 388–398 (1979).
- [3] G. Hummer, “Electrostatic potential of a homogeneously charged square and cube in two and three dimensions,” *J. Electrostatics* 36, 285–291 (1996).
- [4] S. L. Dvorak, D. G. Dudley, “Computation of the incomplete Lipschitz–Hankel integral $\text{Je}_0(a, z)$,” *J. Comput. Phys.* (1990).
- [5] H. Zhu, S. L. Dvorak, “Accurate and efficient computation of the incomplete Lipschitz–Hankel integrals,” *Radio Science* 40 (2005).
- [6] R. Cicchetti, A. Faraone, “Incomplete Hankel and modified Bessel functions: a class of special functions for electromagnetics,” *IEEE Trans. Antennas Propag.* 52(12), 3373–3389 (2004).
- [7] M. M. Agrest, M. S. Maksimov, *Theory of Incomplete Cylindrical Functions and their Applications*, Springer (1971).